

Axioms for Top Trading Cycles in Multi-Object Reallocation*

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May 20, 2026

Abstract

This paper studies multi-object reallocation without monetary transfers, where agents initially own multiple indivisible objects and have strict preferences over bundles (e.g., shift exchange among workers at a firm). We focus first on marginal rules, which elicit only rankings over individual objects, and characterize the generalized Top Trading Cycles rule (TTC) on the lexicographic and responsive domains. On the lexicographic domain, TTC is the unique balanced rule satisfying Pareto efficiency, the worst-endowment lower bound, and either truncation-proofness or drop strategy-proofness. On the responsive domain, TTC is the unique marginal rule satisfying individual-good efficiency, truncation-proofness, and either the worst-endowment lower bound or individual rationality. In the Shapley–Scarf housing market, TTC is characterized by Pareto efficiency, individual rationality, and truncation-proofness. We also consider conditionally lexicographic preferences, which admit complementarities and are represented by lexicographic preference trees. On this domain, the augmented Top Trading Cycles rule (ATTC) is the unique balanced rule satisfying Pareto efficiency, the worst-endowment lower bound, and drop strategy-proofness. The conditionally lexicographic preferences are a maximal domain for the equivalence between Pareto efficiency and individual-good efficiency, and also for the compatibility of balancedness, Pareto efficiency, and individual rationality.

Keywords: Top Trading Cycles; multi-object exchange; responsive preferences; lexicographic preferences; truncation-proofness; worst-endowment lower bound.

JEL Classification: C78; D47; D71.

*We are especially grateful to Ivan Balbuzanov for his guidance and support throughout this project. We also thank Aelya Altuntař, Onur Kesten, Bettina Klaus, Alexandru Nichifor, Szilvia Papai, Tom Wilkening, and Steven Williams, as well as seminar participants at HEC Lausanne, Deakin University, the 2024 Economic Theory Festival at UNSW, ESAM 2024, and MATCH-UP 2024 for their helpful comments and suggestions. Feng gratefully acknowledges financial support from the National Natural Science Foundation of China No. 72503028. Coreno received financial support from an Australian Government RTP Scholarship, a University of Melbourne FBE-GRATS Scholarship, and an M. A. Bartlett Scholarship. Coreno and Feng gratefully acknowledge financial support from the Swiss National Science Foundation (SNSF) through Project 100018-231777. This article is based upon work supported by the National Science Foundation under Grant No. DMS-1928930 and by the Alfred P. Sloan Foundation under grant G-2021-16778, while authors were in residence at the Simons Laufer Mathematical Sciences Institute (formerly MSRI), Berkeley, CA, Fall 2023.

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1 Introduction

We study multi-object reallocation without monetary transfers. A problem instance consists of a group of agents, an initial endowment in which each agent owns a set of heterogeneous and indivisible objects, and each agent’s strict preferences over bundles. A rule maps reported preferences to a feasible reassignment. When each agent is endowed with a single object, the model reduces to the classic housing market of [Shapley and Scarf \(1974\)](#).

Reallocation problems arise in a variety of applications. Firms plan shift schedules months in advance, and employees exchange their assigned shifts (e.g., [Manjunath and Westkamp \(2021\)](#)). Universities exchange students through programs such as the Tuition Exchange Program in the United States and Erasmus in Europe (e.g., [Dur and Ünver \(2019\)](#); [Bloch et al. \(2020\)](#)). Living-donor organ exchange programs provide further examples featuring both single- and multi-object exchange (e.g., [Roth et al. \(2004, 2005\)](#); [Ergin et al. \(2017, 2020\)](#)).¹

Multi-object reallocation confronts two difficulties that are largely absent in single-object reallocation. The first is the vast number of possible bundles, which makes it infeasible for agents to accurately report their preferences over bundles in practice.² As [Roth \(2015, p. 331\)](#) explains, “a practical mechanism must simplify the language in which preferences can be reported.” Accordingly, we focus primarily on *marginal* rules (e.g., [Manjunath and Westkamp \(2025\)](#); [Biró et al. \(2022a\)](#)), under which agents report only rankings over individual objects (their *marginal preferences*).³

We study two preference domains on which marginal preferences are informative about bundle comparisons. On the lexicographic domain, an agent’s ranking over individual objects fully determines her preferences over bundles, so marginal rules are without loss of generality. On the more general responsive domain ([Roth \(1985\)](#)), marginal preferences remain informative—especially for bundles that differ by one object—though they need not determine all bundle comparisons. Both domains rule out complementarities; to capture these, [Section 6](#) considers a domain with a rich but succinct reporting language based on “lexicographic preference trees”.

The second difficulty is the tension among efficiency, voluntary participation, and incentive compatibility. The ideal desiderata—*Pareto efficiency (PE)*, *individual rationality (IR)*, and *strategy-proofness (SP)*—are jointly attainable in housing markets,⁴ but they become incompatible once some agent owns more than one object ([Sönmez \(1999\)](#)). Insisting on strategy-proofness

¹Related work studies course allocation as a *multi-object assignment* problem without endowments (e.g., [Budish \(2011\)](#); [Budish and Cantillon \(2012\)](#); [Bichler et al. \(2021\)](#)).

²For instance, in shift reallocation with only 20 shifts, there are $\binom{20}{5} = 15,504$ distinct five-shift bundles.

³This reporting format is standard in practice: in the National Resident Matching Program (NRMP), for example, hospitals submit rank-order lists over individual doctors despite holding potentially complex preferences over sets of doctors (e.g., [Roth and Peranson \(1999\)](#), [Milgrom \(2009, 2011\)](#)).

⁴In housing markets, these properties uniquely characterize the Top Trading Cycles rule ([Ma \(1994\)](#)).

forces major concessions: under mild auxiliary axioms, combining SP with PE leads to Sequential Dictatorships (which effectively ignore the initial ownership structure; see Pápai (2001); Klaus and Miyagawa (2002); Ehlers and Klaus (2003); Hatfield (2009)), and combining SP with IR yields Segmented Trading Cycles rules (which permit severe inefficiency; see Pápai (2003, 2007)).

In this paper, we circumvent the incompatibility by making modest compromises relative to the ideal notions of efficiency and incentive compatibility. We study the *generalized Top Trading Cycles rule (TTC)*, a multi-object extension of the celebrated Top Trading Cycles algorithm (attributed to David Gale by Shapley and Scarf (1974)). We show that TTC is the unique rule satisfying natural counterparts to PE, IR, and SP across a range of preference domains. More concretely, we characterize TTC using *individual-good efficiency* (IGE, a tractable relaxation of PE), the participation guarantees IR and the *worst-endowment lower bound* (WELB, an object-wise guarantee), and the incentive requirements *truncation-proofness* (TP) and *drop strategy-proofness* (DSP).

On the lexicographic domain, TTC is the only balanced rule satisfying PE, WELB, and either TP or DSP (Theorems 1 and 2). On the responsive domain, where no marginal rule can satisfy PE and IR simultaneously (Manjunath and Westkamp (2025)), TTC is the unique marginal rule satisfying IGE, TP, and either WELB or IR (Theorems 3 and 4). Moreover, no marginal rule on the responsive domain satisfies IGE, IR, and DSP (Corollary 1). In housing markets, our axioms simplify and TTC is characterized by PE, IR, and TP (Theorem 5). We briefly discuss these properties below, deferring formal definitions and further intuition to Section 2.2.

Balanced rules assign each agent the same number of objects she initially owns. In many applications this matches a feasibility constraint imposed directly on the set of allocations.

Individual rationality (IR) is the familiar bundle-level requirement that no agent is worse off than at her endowment. The worst-endowment lower bound (WELB) is an object-wise guarantee that ensures no agent receives any individual object that she ranks below the worst in her endowment according to her marginal preferences. IR and WELB coincide in single-object reallocation. In multi-object settings the properties diverge, but IR implies WELB under mild additional structure.⁵ TTC satisfies both guarantees on each domain we study.

For efficiency, we use Pareto efficiency (PE) when it is attainable and individual-good efficiency (IGE) as a tractable relaxation. IGE relaxes PE by ruling out mutually beneficial exchanges where each participating agent trades a single object. Such trades are comparatively easy to identify and execute, and under responsive preferences they can be detected using only marginal preferences. The two properties coincide in the single-object environment, and in multi-object reallocation with (conditionally) lexicographic preferences. On the responsive

⁵See Lemma 1 and the surrounding discussion.

domain, the relaxation is substantive: realizing Pareto improvements can be computationally demanding (e.g., [De Keijzer et al. \(2009\)](#); [Aziz et al. \(2019\)](#)) and, among marginal rules, PE is incompatible with IR ([Manjunath and Westkamp \(2025\)](#)).

Drop strategy-proofness (DSP) and truncation-proofness (TP) protect against simple manipulations of agents’ marginal preferences. An agent implements a *drop strategy* by demoting a single object she does not own to the bottom of her marginal preference list ([Altuntaş et al. \(2023\)](#)); she implements a *truncation strategy* by demoting every unowned object ranked below a chosen cutoff. Intuitively, truncation strategies can be thought of as “shortening the list of acceptable objects” in the standard sense from two-sided matching (e.g., [Mongell and Roth \(1991\)](#); [Roth and Rothblum \(1999\)](#); [Ehlers \(2008\)](#)). DSP and TP require that no agent can improve her assignment using the corresponding class of manipulations.

In addition to marginal preferences, we also consider a richer reporting language that can express complementarities. We study *conditionally lexicographic* preferences, which admit a compact representation via “lexicographic preference trees” and preserve the equivalence between IGE and PE found under lexicographic preferences. On this domain, we study the *Augmented Top Trading Cycles rule (ATTC)* introduced by [Fujita et al. \(2018\)](#). We show that ATTC satisfies natural extensions of our key properties and is the unique balanced rule satisfying PE, WELB, and DSP (Theorem 6). Finally, we establish two maximal-domain results. The conditionally lexicographic preferences form a maximal domain on which PE and IGE are equivalent (Proposition 5); they also form a maximal domain on which BAL, PE, and IR are compatible (Proposition 6).

The paper proceeds as follows. Section 2 introduces the model, preference domains, properties, and TTC. Sections 3 and 4 present our characterizations on the lexicographic and responsive domains, and Section 5 specializes the results to housing markets. Section 6 studies conditionally lexicographic preferences and characterizes ATTC. Section 7 discusses related literature. The appendices formalize lexicographic preference trees, contain omitted proofs, and provide additional examples.

2 Model

Let $N = \{1, \dots, n\}$ be a finite set of $n \geq 2$ *agents*. Let O be a finite set of heterogeneous and indivisible *objects* such that $|O| \geq n$. A *bundle* is a subset of O . Let 2^O denote the set of bundles. We denote generic elements of O by lowercase letters (e.g., x, y, z), and generic elements of 2^O by uppercase letters (e.g., X, Y, Z). To simplify notation, when there is no risk of confusion, we identify a singleton set $\{x\}$ with the element x itself. For example, we write $X \cup x$ to denote $X \cup \{x\}$.

An *allocation* is a function $\mu : N \rightarrow 2^O$ such that (i) for all $i \in N$, $\mu(i) \neq \emptyset$, (ii) for all

$i, j \in N$, $i \neq j$ implies $\mu(i) \cap \mu(j) = \emptyset$, and (iii) $\bigcup_{i \in N} \mu(i) = O$. Thus, an allocation μ can be represented as a profile $(\mu_i)_{i \in N}$ of nonempty, pairwise disjoint bundles satisfying $\bigcup_{i \in N} \mu_i = O$. For each $i \in N$, μ_i is called agent i 's *assignment* at μ . Let $\mathcal{A}$ denote the set of allocations.

The *initial allocation*, also referred to as the *endowment allocation*, is denoted by $\omega = (\omega_i)_{i \in N}$. For each $i \in N$, ω_i is called agent i 's *endowment*. The initial *owner* of an object o is the unique agent i with $o \in \omega_i$.

Each agent i has a (*strict*) *preference* relation P_i on the set of bundles. We assume that P_i belongs to some specified subset $\mathcal{P}_i$ of all strict preference relations on 2^O . In subsequent sections, we impose further structure on the sets $\mathcal{P}_i$. If agent i prefers bundle X to Y , then we write $X P_i Y$. Let R_i denote the “at least as good as” relation associated with P_i , defined by $X R_i Y$ if and only if $(X P_i Y \text{ or } X = Y)$.⁶ Given a nonempty bundle $X \in 2^O$, $\max_{P_i}(X)$ denotes the most-preferred object in X at P_i , i.e., $\max_{P_i}(X) = x$ if $x \in X$ and $x R_i y$ for all $y \in X$. Similarly, $\min_{P_i}(X)$ denotes the least-preferred object in X at P_i , i.e., $\min_{P_i}(X) = x$ if $x \in X$ and $y R_i x$ for all $y \in X$. A *preference profile* is an indexed family $P = (P_i)_{i \in N}$ of preference relations. The *domain* is the product set $\mathcal{P} := \prod_{i \in N} \mathcal{P}_i$, representing all possible preference profiles under consideration.

An object reallocation problem (or simply a *problem*) is a triple (N, ω, P) . Since (N, ω) remains fixed throughout, we identify a problem with its preference profile P . Thus, the domain $\mathcal{P} = \prod_{i \in N} \mathcal{P}_i$ of preference profiles represents the set of all problems.

A *rule* (on $\mathcal{P}$) is a function $\varphi : \mathcal{P} \rightarrow \mathcal{A}$ that associates with each preference profile P an allocation $\varphi(P)$. For each $i \in N$, $\varphi_i(P)$ denotes agent i 's assignment at $\varphi(P)$.

Marginal rules

A rule is called *marginal* if it can be implemented with a simple reporting language consisting of linear orders over individual objects. Marginal rules are attractive in applications because they ask agents only for a ranking over individual objects, rather than a ranking over all feasible bundles, and thus substantially reduce the informational and cognitive burden on participants. A prominent example is the National Resident Matching Program (NRMP), in which hospitals submit rank-order lists over individual doctors, even though they may have complex preferences over sets of doctors (see, e.g., Roth and Peranson (1999), Roth (2002), Milgrom (2009, 2011)).

Formally, given a preference relation P_i on 2^O , the *marginal preference* over a subset $X \subseteq O$ of individual objects, denoted $P_i|_X$, is the restriction of P_i to singleton subsets of X .⁷ Similarly, $R_i|_X$ denotes the restriction of R_i to singleton subsets of X . We often represent a marginal preference $P_i|_X$ as an ordered list of objects; for example, $P_i|_X : x_1, x_2, \dots, x_{|X|}$ means that

⁶Formally, R_i is a linear order (i.e., a *complete*, *transitive*, and *antisymmetric* binary relation) on 2^O , and P_i is the strict (i.e., *irreflexive* and *asymmetric*) part of R_i .

⁷Equivalently, $P_i|_X$ is the strict linear order on X such that, for all $x, y \in X$, $x P_i|_X y$ if and only if $x P_i y$.

$x_1 P_i x_2 P_i \cdots P_i x_{|X|}$, and $P_i|_X : x_1, x_2, \dots, x_k, \dots$ means that $x_1 P_i x_2 P_i \cdots P_i x_k P_i o$ for all $o \in X \setminus \{x_1, x_2, \dots, x_k\}$. Given a preference profile $P = (P_i)_{i \in N}$, the *marginal preference profile* over X is the profile $P|_X = (P_i|_X)_{i \in N}$.

A rule is marginal if it depends solely on agents' marginal preferences over O .

Definition 1. A rule φ is *marginal (MAR)* if, for all $P, P' \in \mathcal{P}$, $P|_O = P'|_O$ implies $\varphi(P) = \varphi(P')$.

Marginal rules are most natural in settings where objects behave like substitutes, so that agents' preferences over bundles are captured reasonably well by their rankings over individual objects. Most of our analysis focuses on these settings (Sections 3–5). The main limitation of marginal rules is that they do not allow agents to express complementarities across objects. To address complementarities, Section 6 (and Appendix A) considers a richer but still succinct reporting language based on “lexicographic preference trees”.

2.1 Preference domains

We now describe the lexicographic and responsive domains, two structured preference domains where objects behave like substitutes.

An agent has lexicographic preferences if, when evaluating distinct bundles X and Y , she prefers the bundle containing the most-preferred object in $X \cup Y$; if the most-preferred object in $X \cup Y$ is common to X and Y , then she prefers the bundle containing the second-most-preferred object in $X \cup Y$, and so on.⁸ Formally, a preference relation P_i on 2^O is *lexicographic* if, for any two distinct bundles X and Y ,

$$X P_i Y \iff \max_{P_i} (X \triangle Y) \in X,$$

where $X \triangle Y = (X \setminus Y) \cup (Y \setminus X)$ is the symmetric difference between X and Y . For each $i \in N$, let $\mathcal{L}_i$ be the set of lexicographic preferences on 2^O , and let $\mathcal{L} := \prod_{i \in N} \mathcal{L}_i$ be the *lexicographic domain*.

Although it is somewhat restrictive, the lexicographic domain is a natural starting point in our analysis. Each lexicographic preference $P_i \in \mathcal{L}_i$ is uniquely determined by its marginal preference $P_i|_O$ over individual objects; that is, for all $P_i, P'_i \in \mathcal{L}_i$, $P_i|_O = P'_i|_O$ if and only if $P_i = P'_i$. Consequently, any rule defined on $\mathcal{L}$ is automatically marginal. We therefore identify each $P_i \in \mathcal{L}_i$ with its associated marginal preference $P_i|_O$ and write $P_i : x_1, x_2, \dots, x_{|O|}$ if $x_1 P_i x_2 P_i \cdots P_i x_{|O|}$.

Responsiveness is a more general condition first studied by Roth (1985) for many-to-one matching models. An agent has responsive preferences if, for any two bundles that differ in

⁸Similar “take the best” decision heuristics have been documented in psychology (e.g., Gigerenzer and Goldstein (1996), Gigerenzer and Todd (1999)).

one object, she prefers the bundle containing the more-preferred object. Formally, a preference relation P_i on 2^O is *responsive* if, for any bundle X and any objects $y, z \in O \setminus X$,

$$(X \cup y) P_i (X \cup z) \iff y P_i z.$$

Responsive preferences rule out complementarities because the relative ranking between any two objects is independent of the other objects they are obtained with. For each $i \in N$, let $\mathcal{R}_i$ denote the set of responsive preferences on 2^O , and let $\mathcal{R} := \prod_{i \in N} \mathcal{R}_i$ be the *responsive domain*.

It is helpful to compare the lexicographic and responsive domains with the domain of monotonic preferences. A preference relation P_i on 2^O is *monotonic* if, for any bundles X and Y , $X R_i Y$ whenever $X \supseteq Y$. For each $i \in N$, let $\mathcal{M}_i$ be the set of monotonic preferences on 2^O , and let $\mathcal{M} := \prod_{i \in N} \mathcal{M}_i$ be the *monotonic domain*.

Every lexicographic preference is both responsive and monotonic, while responsiveness and monotonicity are logically independent. Formally, for each $i \in N$, we have $\mathcal{L}_i \subseteq \mathcal{R}_i \cap \mathcal{M}_i$ (with strict inclusion if $|O| \geq 3$), $\mathcal{M}_i \not\subseteq \mathcal{R}_i$ (whenever $|O| \geq 3$), and $\mathcal{R}_i \not\subseteq \mathcal{M}_i$.⁹

2.2 Properties of allocations and rules

This section introduces the efficiency, participation, and incentive properties that appear in our characterizations. Unless stated otherwise, these properties are defined for an arbitrary domain $\mathcal{P}$ of strict preferences over bundles. Our main focus will be the lexicographic and responsive domains introduced above.

Our first requirement is that each agent ends up with the same number of objects as initially endowed. An allocation μ is *balanced* if, for each agent $i \in N$, $|\mu_i| = |\omega_i|$. Let $\mathcal{A}^{\text{BAL}}$ denote the set of balanced allocations.

Definition 2. A rule φ satisfies *balancedness* (BAL) if, for each $P \in \mathcal{P}$, $\varphi(P) \in \mathcal{A}^{\text{BAL}}$.

In many applications, balancedness is best viewed as a feasibility constraint on the set of admissible allocations. In shift reallocation, for instance, it prevents overwork or underemployment and may reflect contractual constraints on the number of shifts per worker (Manjunath and Westkamp (2021, 2025)); in student and tuition exchange programs, it helps to maintain reciprocity and control education costs at popular schools (Andersson et al. (2021), Biró et al. (2022a), Dur and Ünver (2019)).¹⁰ As in the aforementioned papers, one could restrict attention from the outset to balanced allocations and omit the balancedness property altogether. For transparency, we model balancedness as a property of rules rather than a model restriction: this

⁹For example, if $O = \{a, b\}$, then $P_i : \{a\}, \emptyset, \{a, b\}, \{b\}$ and $P'_i : \emptyset, \{a\}, \{b\}, \{a, b\}$ are responsive but not monotonic. If $O = \{a, b, c\}$, then $P'_i : \{a, b, c\}, \{a, c\}, \{a, b\}, \{b, c\}, \{a\}, \{b\}, \{c\}, \emptyset$ is monotonic but not responsive.

¹⁰Dur and Ünver (2019) document several cases in which such long-run imbalances led to the failure of an exchange program.

allows us to keep the logical relationships between balancedness and other properties visible (see Lemma 1, Lemma 3, and Example 8).

2.2.1 Efficiency

An allocation $\bar{\mu}$ *Pareto dominates* another allocation μ at a preference profile P if (i) for all $i \in N$, $\bar{\mu}_i R_i \mu_i$, and (ii) for some $i \in N$, $\bar{\mu}_i P_i \mu_i$. An allocation μ is *Pareto efficient* at P if it is not Pareto dominated at P by any allocation.

Definition 3. A rule φ satisfies *Pareto efficiency (PE)* if, for each $P \in \mathcal{P}$, $\varphi(P)$ is Pareto efficient at P .

In principle, any allocation that violates Pareto efficiency could be destabilized by a coalition of agents carrying out a mutually beneficial exchange. In practice, however, such exchanges may be hard to realize. A coalition would typically need detailed information about the agents' preferences over bundles and a way to coordinate intricate trades of multiple objects among many agents. Even with the required preference information, finding a Pareto improvement—or certifying that none exists—is computationally demanding. In particular, under additive preferences (a subclass of responsive preferences), deciding whether a Pareto improvement exists is NP-complete (e.g., De Keijzer et al. (2009); Aziz et al. (2019)).

Individual-good efficiency relaxes PE by ruling out a restricted class of mutually beneficial exchanges—multilateral trades where each participating agent offers a single object in return for another. Formally, a *trading cycle* at an allocation μ is a cyclic sequence

$$C = (o_1, i_1, o_2, i_2, \dots, i_{k-1}, o_k, i_k, o_{k+1} = o_1)$$

consisting of $k \geq 1$ distinct objects and k distinct agents such that, for each $\ell \in \{1, \dots, k\}$, $o_\ell \in \mu_{i_\ell}$. For each $\ell \in \{1, \dots, k\}$, we interpret o_ℓ as the object relinquished by agent i_ℓ and $o_{\ell+1}$ as the object received. We say that C is *Pareto improving* at a preference profile P if

$$\text{for each } \ell \in \{1, \dots, k\}, \quad ((\mu_{i_\ell} \setminus \{o_\ell\}) \cup \{o_{\ell+1}\}) P_{i_\ell} \mu_{i_\ell}.$$

An allocation μ is *individual-good efficient* at P if no trading cycle at μ is Pareto improving at P .¹¹

Definition 4. A rule φ satisfies *individual-good efficiency (IGE)* if, for each $P \in \mathcal{P}$, $\varphi(P)$ satisfies individual-good efficiency at P .

Although PE and IGE coincide on the lexicographic domain (Aziz et al. (2019)), on more general preference domains an allocation may satisfy IGE even when further Pareto improvements

¹¹This terminology is borrowed from Biró et al. (2022a). Similar properties are studied in Aziz et al. (2019), Caspari (2020), and Coreno and Balbuzanov (2026).

are possible through more complex trades. This loss of efficiency is the price we pay for a notion that is tractable and achievable in practice. IGE focuses on single-object exchanges, which are relatively easy to organize,¹² and under responsive preferences such exchanges can be detected using only the information elicited by a marginal rule. In addition, there are polynomial-time algorithms for deciding whether a given allocation satisfies IGE on each of the domains we consider (e.g., [Cechlárová et al. \(2014\)](#); [Aziz et al. \(2019\)](#); [Fujita et al. \(2018\)](#)).

2.2.2 Participation guarantees

An allocation μ is *individually rational* at a preference profile P if, for each $i \in N$, $\mu_i R_i \omega_i$.

Definition 5. A rule φ satisfies *individual rationality (IR)* if, for each $P \in \mathcal{P}$, $\varphi(P)$ is individually rational at P .

Individual rationality is the standard participation guarantee: at the outcome, no agent is worse off than under her endowment. IR is stated in terms of preferences over bundles and thus requires more information than a marginal rule generally elicits. We pair it with an object-wise participation guarantee expressed purely in terms of marginal preferences: no agent receives any individual object that she ranks below her least-preferred endowment object. Formally, an allocation μ satisfies the *worst-endowment lower bound* at a preference profile P if, for each $i \in N$ and each $o \in \mu_i$, $o R_i \min_{P_i}(\omega_i)$.

Definition 6. A rule φ satisfies the *worst-endowment lower bound (WELB)* if, for each $P \in \mathcal{P}$, $\varphi(P)$ satisfies the worst-endowment lower bound at P .

WELB coincides with IR in the single-object environment, where each agent owns and receives exactly one object. In multi-object problems the two properties diverge: IR permits an agent to receive an individually unattractive object as part of a sufficiently desirable bundle, whereas WELB rules out such objects altogether. A useful interpretation is that WELB gives each agent veto power over any individual object owned by others. By ranking some object o below every object she owns, an agent can guarantee that she will never be assigned o .

The two guarantees are closely related under additional structure. On the responsive domain, marginality and IR jointly imply WELB (and BAL).

Lemma 1. *On the responsive domain, if a marginal rule satisfies IR, then it satisfies BAL and WELB.*

A similar implication holds on any domain in which bundles violating WELB are treated as unacceptable (e.g., [Andersson et al. \(2021\)](#), [Biró et al. \(2022a\)](#)).¹³ We do not impose this

¹²A similar relaxation is “pair efficiency” ([Ekici \(2024\)](#)), which precludes only mutually beneficial *bilateral* trades.

¹³Formally, consider a domain $\mathcal{P}^*$ such that, for each $i \in N$ and each $P_i \in \mathcal{P}_i^*$, any bundle that intersects $\{o \in O \mid \min_{P_i}(\omega_i) P_i o\}$ is worse than any bundle that does not. Clearly, any IR rule on $\mathcal{P}^*$ also satisfies WELB.

additional restriction; instead we treat IR and WELB as independent properties in order to separate their respective roles in the results.

2.2.3 Incentive properties

Given a preference profile $P = (P_i)_{i \in N}$ and a preference relation P'_i for agent i , we denote by (P'_i, P_{-i}) the preference profile in which agent i reports P'_i and every other agent $j \in N \setminus \{i\}$ reports P_j . Given a rule φ , we say that agent i can *manipulate* φ at P by reporting P'_i if $\varphi_i(P'_i, P_{-i}) P_i \varphi_i(P)$. A rule is *strategy-proof* if no agent can manipulate it by reporting any preference relation.

Definition 7. A rule φ satisfies *strategy-proofness (SP)* if, for each $P \in \mathcal{P}$, each $i \in N$, and each $P'_i \in \mathcal{P}_i$, $\varphi_i(P) R_i \varphi_i(P'_i, P_{-i})$.

Because PE, IR, and SP are incompatible once some agent owns more than one object (Sönmez (1999), Todo et al. (2014)), we will not insist on full SP. Instead, we relax SP by restricting the set of manipulation strategies under consideration. In particular, we focus on simple manipulations—subset-drop, drop, and truncation strategies—that act directly on agents' marginal preferences over individual objects and are especially salient when studying marginal rules (see, e.g., Biró et al. (2022a,b), Altuntaş et al. (2023)).

Subset-drop, drop, and truncation strategies. Fix an agent i and a preference relation $P_i \in \mathcal{P}_i$. A subset-drop strategy for P_i is a report whose marginal preference is obtained from $P_i|_O$ by “dropping” some subset $X \subseteq O \setminus \omega_i$ to the bottom, preserving the relative order within X and within $O \setminus X$. Any preference $P'_i \in \mathcal{P}_i$ with this marginal preference is called a subset-drop strategy for P_i . Formally, $P'_i \in \mathcal{P}_i$ is a *subset-drop strategy* for P_i if there exists $X \subseteq O \setminus \omega_i$ such that

- (i) $P_i|_X = P'_i|_X$,
- (ii) $P_i|_{O \setminus X} = P'_i|_{O \setminus X}$, and
- (iii) for all $x \in X$ and all $y \in O \setminus X$, $y P'_i x$.

In this case, we say that P'_i (respectively $P'_i|_O$) is obtained from P_i (respectively $P_i|_O$) by *dropping* X . Let $\mathcal{S}_i(P_i)$ denote the set of all subset-drop strategies for P_i .

Within the class of subset-drop strategies, we distinguish two important subclasses. A subset-drop strategy P'_i is called a *drop strategy* for P_i if it is obtained by dropping a singleton subset $\{x\} \subseteq O \setminus \omega_i$, i.e., by moving a single object $x \in O \setminus \omega_i$ to the bottom of the marginal preferences. Let $\mathcal{D}_i(P_i)$ denote the set of all drop strategies for P_i .

A subset-drop strategy P'_i is called a *truncation strategy* for P_i if it is obtained by dropping a *tail subset* of the form $\{o \in O \setminus \omega_i \mid x R_i o\}$ for some $x \in O$, i.e., by moving all objects that are

Furthermore, if preferences P_i are assumed to be lexicographic (or responsive) when restricted to “acceptable” bundles (subsets of $\{o \in O \mid o R_i \min_{P_i}(\omega_i)\}$), then our characterization results go through unchanged on $\mathcal{P}^*$.

weakly worse than some object x to the bottom of the marginal preferences. Let $\mathcal{T}_i(P_i)$ denote the set of all truncation strategies for P_i .

It is convenient to single out those truncation strategies with a specified cutoff object. A truncation strategy P'_i is called a *truncation of P_i at y* if it is obtained by dropping the strict tail subset $\{o \in O \setminus \omega_i \mid y P_i o\}$.¹⁴ In this case, there is no $z \in O \setminus \omega_i$ such that $y P'_i z P'_i \min_{P_i}(\omega_i)$.

In the single-object environment (Section 5), where ω_i is a singleton and thus $\omega_i = \min_{P_i}(\omega_i)$, a subset-drop strategy can be interpreted as “declaring some of the other agents’ objects unacceptable”, and a truncation strategy simply “shortens the list of acceptable objects” in the standard sense from two-sided matching (see, e.g., Mongell and Roth (1991); Roth and Rothblum (1999); Ehlers (2008)). This intuition extends to the multi-object environment, though objects ranked below $\min_{P_i}(\omega_i)$ need not be unacceptable in the usual sense.¹⁵ Nevertheless, under WELB, a subset-drop strategy effectively vetoes some of the other agents’ objects: no rule satisfying WELB ever assigns to agent i any object ranked below $\min_{P_i}(\omega_i)$.

Definition 8. A rule φ satisfies

- *truncation-proofness (TP)* if, for each $P \in \mathcal{P}$, each $i \in N$, and each $P'_i \in \mathcal{T}_i(P_i)$, we have $\varphi_i(P) R_i \varphi_i(P'_i, P_{-i})$.
- *drop strategy-proofness (DSP)* if, for each $P \in \mathcal{P}$, each $i \in N$, and each $P'_i \in \mathcal{D}_i(P_i)$, we have $\varphi_i(P) R_i \varphi_i(P'_i, P_{-i})$.
- *subset-drop strategy-proofness (SDSP)* if, for each $P \in \mathcal{P}$, each $i \in N$, and each $P'_i \in \mathcal{S}_i(P_i)$, we have $\varphi_i(P) R_i \varphi_i(P'_i, P_{-i})$.

SDSP implies both TP and DSP, as it defends against the largest set of manipulation strategies. TP and DSP are logically independent in general.

Example 1. Let $O = \{a, b, c, d, e, x, y\}$, $\omega_i = \{x, y\}$, and $P_i \in \mathcal{P}_i$ be such that $P_i|_O : a, b, \underline{x}, c, d, \underline{y}, e$ (agent i ’s endowment is underlined for emphasis). Then:

- any $P_i^1 \in \mathcal{P}_i$ with $P_i^1|_O : b, \underline{x}, c, d, \underline{y}, e, a$ is a drop strategy for P_i , obtained by dropping object a . Note that P_i^1 is not a truncation strategy for P_i (e.g., the relations disagree about the best object in $O \setminus \omega_i$);
- any $P_i^2, P_i^3 \in \mathcal{P}_i$ with $P_i^2|_O : a, b, \underline{x}, c, \underline{y}, d, e$ and $P_i^3|_O : a, \underline{x}, \underline{y}, b, c, d, e$, are truncation strategies for P_i , obtained by dropping the tail subsets $\{d, e\}$ and $\{b, c, d, e\}$, respectively. Neither P_i^2 nor P_i^3 is a drop strategy for P_i (however, each can be obtained from P_i via a sequence of drop strategies). $\diamond$

¹⁴Equivalently, P'_i is a truncation of P_i at y if (i) $P_i|_{\omega_i} = P'_i|_{\omega_i}$, (ii) $P_i|_{O \setminus \omega_i} = P'_i|_{O \setminus \omega_i}$, and (iii) for each $x \in O \setminus \omega_i$ with $y P_i x$, $\min_{P'_i}(\omega_i) P'_i x$.

¹⁵Our results continue to hold on related domains in which bundles containing objects ranked below $\min_{P_i}(\omega_i)$ are explicitly unacceptable; see footnote 13 and the surrounding discussion.

2.3 Top Trading Cycles

A trading cycle at ω (or simply a *trading cycle*¹⁶) is a cyclic sequence

$$C = (o_1, i_1, o_2, i_2, \dots, i_{k-1}, o_k, i_k, o_{k+1} = o_1)$$

consisting of $k \geq 1$ distinct objects and k distinct agents such that, for each $\ell \in \{1, \dots, k\}$, $o_\ell \in \omega_{i_\ell}$. Each object o_ℓ on C precedes (or “points to”) its owner i_ℓ , and each agent i_ℓ on C precedes (or “points to”) the object $o_{\ell+1}$. The sets of agents and objects on C are denoted by $N(C) = \{i_1, i_2, \dots, i_k\}$ and $O(C) = \{o_1, o_2, \dots, o_k\}$, respectively. An allocation μ is said to *execute* the trading cycle C if it assigns to each agent in $N(C)$ the object she points to on C ; that is, for each $i_\ell \in N(C)$, $o_{\ell+1} \in \mu_{i_\ell}$.

We study a multi-object extension of Gale’s Top Trading Cycles algorithm (Shapley and Scarf (1974)), which determines an allocation by executing a sequence of carefully chosen trading cycles at ω . At each step of this procedure, every agent points to her most-preferred unassigned object, and every unassigned object points to its owner. The resulting directed graph contains at least one trading cycle, and each agent involved in a trading cycle is assigned the object to which she points. All objects involved in a trading cycle are then removed. If unassigned objects remain, then the procedure continues to the next step; otherwise, it terminates with the resulting allocation.

We now formalize the (*generalized*) *Top Trading Cycles rule (TTC)*, which we denote by φ^{TTC} . Given a preference profile $P \in \mathcal{P}$, the allocation $\varphi^{\text{TTC}}(P)$ is determined by running the following (*generalized*) *TTC algorithm* at P . We denote this specific instance as $\text{TTC}(P)$.

Algorithm: $\text{TTC}(P)$

Initialization: Set $\mu^0 := (\emptyset)_{i \in N}$ and $O^1 := O$.

Step $t \geq 1$: Each agent i points to $\max_{P_i}(O^t)$, and each object in O^t points to its owner. Let $\mathcal{C}_t(P)$ be the resulting set of trading cycles. For each $C \in \mathcal{C}_t(P)$, assign to each $i \in N(C)$ the object she points to in C ; this yields the partial allocation μ^t with

$$\mu_i^t = \begin{cases} \mu_i^{t-1} \cup \{\max_{P_i}(O^t)\}, & \text{if } i \in \bigcup_{C \in \mathcal{C}_t(P)} N(C), \\ \mu_i^{t-1}, & \text{otherwise.} \end{cases}$$

Remove all objects involved in trading cycles to form the set $O^{t+1} = O^t \setminus \bigcup_{C \in \mathcal{C}_t(P)} O(C)$.

Proceed to step $t + 1$ if $O^{t+1} \neq \emptyset$; proceed to Termination otherwise.

Termination: Because at least one object is removed at each step, the algorithm terminates at

¹⁶That is, a *trading cycle* refers to a trading cycle at the particular allocation $\mu = \omega$ (as defined in Section 2.2.1).

some step T . Return the allocation $\varphi^{\text{TTC}}(P) := \mu^T$.

Example 2 illustrates the TTC algorithm with a simple three-agent problem.

Example 2. Suppose $N = \{1, 2, 3\}$, $O = \{a, b, c, d\}$, and $\omega = (\{a, b\}, \{c\}, \{d\})$. Consider a preference profile P with marginal preferences $P_1|_O : c, a, d, b$, $P_2|_O : a, b, c, d$, and $P_3|_O : a, c, b, d$. The algorithm $\text{TTC}(P)$ operates as follows.

Step 1: Each agent points to her most-preferred object in O : agent 1 points to c , while agents 2 and 3 point to a . Each object in O points to its owner. There is a trading cycle $C_1(P) = (c, 2, a, 1, c)$, which is executed to yield partial allocation $\mu^1 = (\{c\}, \{a\}, \emptyset)$ and the remaining objects $O^2 = \{b, d\}$.

Step 2: Each agent points to her most-preferred object in $O^2 = \{b, d\}$: agent 1 points to d , while agents 2 and 3 point to b . Each object in O^2 points to its owner. There is a trading cycle $C_2(P) = (d, 3, b, 1, d)$, which is executed to yield the final allocation $\mu^2 = (\{c, d\}, \{a\}, \{b\})$ and the remaining objects $O^3 = \emptyset$.

The algorithm therefore terminates with the allocation $\varphi^{\text{TTC}}(P) = (\{c, d\}, \{a\}, \{b\})$. $\diamond$

The following fact is immediate from the definition of φ^{TTC} .

Fact 1. *On any domain of strict preferences, φ^{TTC} is marginal and satisfies BAL and WELB.*

Beyond these basic properties, the behavior of φ^{TTC} depends on the preference domain.

3 Lexicographic preferences

The lexicographic domain $\mathcal{L}$ is a natural starting point for our analysis because of its tractability. As discussed in Sections 2.1–2.2, this domain has two convenient features. First, each $P_i \in \mathcal{L}_i$ is uniquely determined by its marginal preference $P_i|_O$, so any rule defined on $\mathcal{L}$ is automatically marginal. Second, on $\mathcal{L}$, PE coincides with IGE (see Proposition 3).

On the lexicographic domain, φ^{TTC} behaves especially well. Fujita et al. (2018) show that it always selects an allocation in the core of the associated exchange economy, and hence it satisfies both PE and IR. Although φ^{TTC} necessarily fails SP,¹⁷ Altuntaş et al. (2023) prove that it satisfies SDSP; thus it also satisfies TP and DSP.¹⁸ These observations yield the following fact.

Fact 2. *On the lexicographic domain, φ^{TTC} satisfies PE, IR, and SDSP.*

Together, Facts 1 and 2 show that φ^{TTC} is a balanced rule satisfying PE, WELB, and TP. Among balanced rules, these properties characterize φ^{TTC} on $\mathcal{L}$.

¹⁷PE, IR, and SP are incompatible even for lexicographic preferences (Todo et al. (2014)).

¹⁸Altuntaş et al. (2023) refer to SDSP as “subset total drop strategy-proofness”. Their notion of drop strategy-proofness is stronger than ours, as it also precludes manipulations obtained by dropping objects an agent already owns. Throughout, we use the weaker DSP as it suffices for our characterization results.

Theorem 1. *On the lexicographic domain, a balanced rule satisfies PE, WELB, and TP if and only if it equals φ^{TTC} .*

The proof proceeds by minimal counterexample, using a novel minimality criterion that combines a “size” function (Sethuraman (2016)) and a “similarity” function (Ekici (2024)). Suppose φ satisfies the properties but differs from φ^{TTC} . Call $P \in \mathcal{L}$ a *conflict profile* if $\varphi(P) \neq \varphi^{\text{TTC}}(P)$. For each conflict profile P , define its similarity $\rho(P)$ as the earliest step t of $\text{TTC}(P)$ such that $\varphi(P)$ does not execute all trading cycles in $\mathcal{C}_t(P)$. Among all conflict profiles that minimize ρ , choose a profile that further minimizes the size function $s(P) := \sum_{i \in N} |\{o \in O \mid o R_i \min_{P_i}(\omega_i)\}|$. At the chosen conflict profile P , the deviation from $\varphi^{\text{TTC}}(P)$ forces a Pareto-improving trading cycle at $\varphi(P)$, contradicting IGE (see Appendix B).

Theorem 1 characterizes φ^{TTC} using TP as its incentive requirement. Our next theorem shows that TP can be replaced with the alternative requirement DSP. The key observation is that DSP becomes strictly stronger than TP when imposed together with WELB.¹⁹

Lemma 2. *On the lexicographic domain, if a rule satisfies DSP and WELB, then it satisfies SDSP.*

Theorem 2 refines two characterizations provided by Altuntaş et al. (2023, Theorems 1 and 3).

Theorem 2. *On the lexicographic domain, a balanced rule satisfies PE, WELB, and DSP if and only if it equals φ^{TTC} .*

Proof. By Facts 1 and 2, φ^{TTC} satisfies the stated properties. Conversely, if φ is a rule satisfying BAL, IGE, WELB, and DSP, then Lemma 2 implies that φ satisfies SDSP and hence TP. Therefore, φ satisfies the properties in Theorem 1, which yields $\varphi = \varphi^{\text{TTC}}$. $\square$

Neither of Theorems 1 or 2 remains valid if WELB is replaced with IR.²⁰ Example 9 exhibits a rule $\varphi^{-\text{WELB}}$ on $\mathcal{L}$ that satisfies BAL, PE, IR, TP, and DSP yet differs from φ^{TTC} . The construction exploits exactly the gap between IR and WELB discussed in Section 2.2: under $\varphi^{-\text{WELB}}$, an agent may receive a bundle preferred to her endowment that nevertheless contains a very undesirable object—an outcome forbidden under WELB. The analogous construction is not possible for marginal rules on the responsive domain; in that setting, IR implies WELB and hence suffices for the characterization (Lemma 1 and Theorem 4).²¹

¹⁹TP is not implied by any one of DSP and WELB. Conversely, DSP is not implied by any proper subset of the properties in Theorem 1. The reason is that truncation strategies preserve the relative ranking of unowned objects, while drop strategies can be composed to reorder unowned objects arbitrarily.

²⁰Of course, IR could be added to the list of properties together with WELB but it is implied by the four properties in each theorem.

²¹Similarly, this construction is not possible on the domain $\mathcal{P}^*$ described in footnote 13, as IR implies WELB on that domain.

More generally, Appendix C provides several alternative rules that demonstrate the independence of our properties. Table 1 summarizes the properties satisfied by these rules on the lexicographic domain; it shows that each of the properties in Theorems 1 and 2 is indispensable.

Table 1: Properties of selected rules on $\mathcal{L}$

Rule	BAL	PE	WELB	TP	DSP	IR	SP
φ^{TTC}	✓	✓	✓	✓	✓	✓	
No-trade rule (Ex. 5)	✓		✓	✓	✓	✓	✓
Balanced Serial Dictatorship (Ex. 6)	✓	✓		✓	✓		✓
$\varphi^{-\text{TP}}$ (Ex. 7)	✓	✓	✓			✓	
$\varphi^{-\text{BAL}}$ (Ex. 8)		✓	✓	✓	✓	✓	
$\varphi^{-\text{WELB}}$ (Ex. 9)	✓	✓		✓	✓	✓	

4 Responsive preferences

On the responsive domain, the tension among efficiency, individual rationality, and incentive compatibility becomes sharp. This tension is exacerbated by marginality: Manjunath and Westkamp (2025) show that no marginal rule can simultaneously satisfy even PE and IR. Any characterization on this domain must therefore relax marginality, PE, or IR. We retain marginality because preferences over bundles are difficult to elicit reliably in practice, and we retain IR because it is the standard voluntary-participation requirement. We adopt IGE as a tractable surrogate for PE.²²

This relaxation is not merely a technical convenience. Under responsive preferences, realizing Pareto improvements often requires coordinating intricate multi-object exchanges among many agents, and even deciding whether a Pareto improvement exists is NP-complete (De Keijzer et al. (2009), Aziz et al. (2019)). IGE focuses on the subclass of Pareto improvements that can be realized through single-object trades.

The following example, adapted from Manjunath and Westkamp (2025), illustrates the incompatibility of PE, IR, and marginality.

Example 3. Let $N = \{1, 2\}$, $O = \{a, b, c, d\}$, and $\omega = (\{a, d\}, \{b, c\})$. Let φ be a marginal rule on $\mathcal{R}$ satisfying IR.

Consider $P \in \mathcal{L}$ with $P_1|_O = P_2|_O : a, b, c, d$. Since φ satisfies IR, $\varphi(P) = \omega$. Marginality forces φ to select ω at every responsive profile with the same marginal preferences. In particular,

²²Rules on $\mathcal{R}$ satisfying PE and SP without IR include the Sequential Dictatorships (e.g., Ehlers and Klaus (2003) and Hatfield (2009)); rules satisfying IR and SP without PE include the Segmented Trading Cycles rules of Pápai (2003). Both classes turn out to be marginal as a consequence of SP combined with other properties. Whether there exists a natural class of non-marginal rules on $\mathcal{R}$ with enough structure to yield meaningful possibility or impossibility results remains open; Example 10 provides one artificial construction.

consider $P' \in \mathcal{R}$ with $P'|_O = P|_O$, in which each agent prefers the other's endowment: $\{b, c\} P'_1 \{a, d\}$ and $\{a, d\} P'_2 \{b, c\}$. Then $\varphi(P') = \varphi(P) = \omega$, but ω is Pareto dominated at P' by the bundle swap $(\{b, c\}, \{a, d\})$. Hence φ fails PE at P' . $\diamond$

Example 3 shows that marginality is a substantive constraint on the responsive domain. It forces φ to discard information that the agents possess about their preferences over bundles, leaving PE unattainable at P' . Yet no trading cycle is Pareto improving at P' : under the common marginal preferences, any single-object exchange would require an agent to relinquish an object for a lower-ranked one.

More generally, φ^{TTC} inherits individual-good efficiency on $\mathcal{R}$ from its Pareto efficiency on $\mathcal{L}$: if a Pareto-improving trading cycle existed at $\varphi^{\text{TTC}}(P)$ for some $P \in \mathcal{R}$, then the same cycle would be Pareto improving at the lexicographic profile with the same marginal preferences, contradicting the Pareto efficiency of φ^{TTC} on $\mathcal{L}$. In addition, Altuntaş et al. (2023) show that $\mathcal{R}$ is a maximal domain on which φ^{TTC} satisfies IR. The next fact collects these properties.

Fact 3. *On the responsive domain, φ^{TTC} satisfies IGE and IR.*

Incentive compatibility becomes more fragile on the responsive domain because extending a rule to a larger domain preserves any existing manipulations and may introduce new ones. Nevertheless, φ^{TTC} remains robust against truncation strategies on $\mathcal{R}$.

Proposition 1. *On the responsive domain, φ^{TTC} satisfies TP.*

Combining Fact 1, Fact 3, and Proposition 1, φ^{TTC} is a balanced rule satisfying IGE, WELB, and TP on $\mathcal{R}$. Among marginal rules, these properties continue to identify φ^{TTC} on $\mathcal{R}$ (cf. Theorem 1). In fact, Theorem 3 below shows that uniqueness holds more generally. The key observation is the following lemma: for marginal rules on $\mathcal{R}$, balancedness is implied by WELB and TP (and hence superfluous).

Lemma 3. *On the responsive domain, if a marginal rule satisfies WELB and TP, then it satisfies BAL.*

Theorem 3. *On the responsive domain, a marginal rule satisfies IGE, WELB, and TP if and only if it equals φ^{TTC} .*

Proof. We have shown that φ^{TTC} satisfies the stated properties. Conversely, let φ be a marginal rule on $\mathcal{R}$ satisfying IGE, WELB, and TP. By Lemma 3, φ also satisfies BAL. Hence Theorem 1 implies that φ coincides with φ^{TTC} on $\mathcal{L}$.

Fix any $P' \in \mathcal{R}$, and let P be the unique lexicographic preference profile with $P|_O = P'|_O$. Since φ and φ^{TTC} are marginal, we have

$$\varphi(P') = \varphi(P) = \varphi^{\text{TTC}}(P) = \varphi^{\text{TTC}}(P').$$

Thus φ coincides with φ^{TTC} on $\mathcal{R}$.²³ $\square$

Unlike on the lexicographic domain, where IR cannot generally replace WELB (see Example 9), on $\mathcal{R}$ any marginal rule satisfying IR also satisfies WELB (Lemma 1). Thus WELB can be replaced by IR in Theorem 3.

Theorem 4. *On the responsive domain, a marginal rule satisfies IGE, IR, and TP if and only if it equals φ^{TTC} .*

Proof. We have shown that φ^{TTC} satisfies the stated properties. Conversely, if φ is a marginal rule satisfying IGE, IR, and TP, then Lemma 1 implies that φ also satisfies WELB. Therefore, φ satisfies the properties in Theorem 3, and hence $\varphi = \varphi^{\text{TTC}}$. $\square$

Theorems 3 and 4 characterize φ^{TTC} on $\mathcal{R}$ using TP as the incentive requirement. The next example shows that φ^{TTC} fails the alternative requirement DSP on $\mathcal{R}$.

Example 4. Let $N = \{1, 2, 3\}$ and $\omega = (\{a, b\}, \{c\}, \{d\})$. Let P_2 and P_3 be lexicographic preferences with $P_2|_O : d, b, c, a$ and $P_3|_O : b, a, c, d$.

Let P_1 be a responsive preference with $P_1|_O : d, b, c, a$ and $\{b, c\} P_1 \{a, d\}$. Then φ^{TTC} assigns $\varphi^{\text{TTC}}(P) = (\{a, d\}, \{c\}, \{b\})$. Now suppose agent 1 drops object d , reporting some P'_1 with the marginal preference $P'_1|_O : b, c, a, d$. Then φ^{TTC} assigns $\varphi^{\text{TTC}}(P'_1, P_{-1}) = (\{b, c\}, \{d\}, \{a\})$. Because $\{b, c\} P_1 \{a, d\}$, agent 1 benefits from the drop strategy. $\diamond$

Example 4 suggests that DSP is a rather restrictive requirement on $\mathcal{R}$. Combined with our lexicographic-domain characterization (Theorem 2) and the fact that IR implies WELB for marginal rules on $\mathcal{R}$ (Lemma 1), this observation yields the following impossibility.

Corollary 1. *On the responsive domain, no marginal rule satisfies IGE, IR, and DSP.*

Proof. Toward contradiction, let φ be a marginal rule on $\mathcal{R}$ satisfying IGE, IR, and DSP. Then φ also satisfies BAL and WELB by Lemma 1. Hence Theorem 2 implies that φ coincides with φ^{TTC} on $\mathcal{L}$, and by marginality it must coincide with φ^{TTC} on $\mathcal{R}$. Thus φ fails DSP, a contradiction. $\square$

Corollary 1 shows that any marginal rule on $\mathcal{R}$ that guarantees IGE and IR is necessarily manipulable—even among drop strategies.²⁴ Nevertheless, φ^{TTC} is *not obviously manipulable* in the sense of Troyan and Morrill (2020). Roughly speaking, although profitable deviations can exist, no deviation is clearly beneficial when the agent compares only the best- and worst-case outcomes that can result from truthful reporting and from the deviation.

²³The argument parallels one used by Biró et al. (2022a) to extend a characterization of the “Circulation Top Trading Cycle” rule from the lexicographic to the responsive domain in a model with homogeneous objects.

²⁴A similar argument yields an analogous impossibility result with IR replaced by BAL and WELB: no marginal rule on $\mathcal{R}$ satisfies IGE, BAL, WELB, and DSP.

For each $i \in N$ and each $P_i \in \mathcal{R}_i$, agent i 's *opportunity set* at P_i under φ^{TTC} is the set $\mathcal{O}_i(P_i) = \{\varphi_i^{\text{TTC}}(P_i, P_{-i}^*) \mid P_{-i}^* \in \mathcal{P}_{-i}\}$ of all bundles that i could be assigned upon reporting P_i . For each nonempty subset $\mathcal{X} \subseteq 2^O$, $B_{P_i}(\mathcal{X})$ denotes the most-preferred bundle in $\mathcal{X}$ according to P_i , i.e., $B_{P_i}(\mathcal{X}) \in \mathcal{X}$ and $B_{P_i}(\mathcal{X}) R_i X$ for each $X \in \mathcal{X}$. Similarly, $W_{P_i}(\mathcal{X})$ denotes the least-preferred bundle in $\mathcal{X}$ according to P_i .

Proposition 2. *On the responsive domain, φ^{TTC} is not obviously manipulable.*

More precisely, for each $P \in \mathcal{R}$, each $i \in N$, and each $P'_i \in \mathcal{R}_i$ with $\varphi_i^{\text{TTC}}(P'_i, P_{-i}) P_i \varphi_i^{\text{TTC}}(P)$,

(i) $B_{P_i}(\mathcal{O}_i(P_i)) R_i B_{P_i}(\mathcal{O}_i(P'_i))$, and

(ii) $W_{P_i}(\mathcal{O}_i(P_i)) R_i W_{P_i}(\mathcal{O}_i(P'_i))$.

Table 2 summarizes the properties satisfied by several alternative rules on $\mathcal{R}$ (see the constructions in Appendix C). It shows that the properties used in Theorems 3 and 4, including marginality, are logically independent.

Table 2: Properties of selected rules on $\mathcal{R}$

Rule	BAL	IGE	WELB	TP	DSP	IR	MAR
φ^{TTC}	✓	✓	✓	✓		✓	✓
No-trade rule (Ex. 5)	✓		✓	✓	✓	✓	✓
Balanced Serial Dictatorship (Ex. 6)	✓	✓		✓	✓		✓
$\varphi^{-\text{TP}}$ (Ex. 7)	✓	✓	✓			✓	✓
$\varphi^{-\text{MAR}}$ (Ex. 10)	✓	✓	✓	✓		✓	

5 The single-object environment

We now specialize our analysis to the classic single-object environment (the “housing market”) introduced by Shapley and Scarf (1974). In this environment our properties simplify, and the multi-object characterization in Theorem 1 collapses to a sharper statement.

We formalize the single-object environment as follows. Assume that $O = \{o_1, \dots, o_n\}$ and that each agent $i \in N$ is endowed with the object $\omega_i = o_i$. Each agent i has strict preferences P_i over individual objects in O , and $\mathcal{P} = \prod_{i \in N} \mathcal{P}_i$ denotes the domain of strict preference profiles.²⁵ An allocation is represented as a bijection $\mu : N \rightarrow O$, where μ_i denotes the object assigned to agent i .

In the single-object environment, every allocation is balanced by definition. Moreover, IR and WELB coincide because each agent must be assigned exactly one object, and PE and IGE coincide because any Pareto improvement can be realized by executing a collection of

²⁵In this section only, we adopt the standard convention that each agent has preferences P_i defined directly on the set of objects O . This is only a notational simplification. If instead we maintained the assumption that each agent has lexicographic preferences on 2^O , then the result below would be an immediate corollary of Theorem 1.

disjoint trading cycles. Truncation and drop strategies—as well as the corresponding incentive properties, TP and DSP—are defined exactly as in Section 2.2.3. In this setting, truncation strategies admit a familiar interpretation from the matching literature: a truncation strategy for P_i simply shortens agent i ’s list of acceptable objects by moving the “outside option” ω_i up in the ranking while preserving the ordering of objects in $O \setminus \omega_i$ (see footnote 14 and the surrounding discussion).

Theorem 5. *In the single-object environment, a rule satisfies PE, IR, and TP if and only if it equals φ^{TTC} .*

The proof closely parallels that of Theorem 1 and is therefore omitted.

Theorem 5 refines several characterizations of φ^{TTC} in housing markets. In particular, Ma (1994) shows that φ^{TTC} is the unique rule satisfying PE, IR, and SP, and Altuntaş et al. (2023) prove that SP can be replaced with either DSP or the weaker “upper invariance”.²⁶ In the single-object environment, upper invariance together with IR implies TP; hence Theorem 5 establishes the same uniqueness under weaker criteria.

The theorem also complements characterizations using relaxed efficiency requirements. For example, Ekici (2024) replaces PE with “pair efficiency” while retaining IR and SP, and Chen et al. (2025) further weaken SP to upper invariance. Theorem 5 does not permit a similar refinement: pair efficiency together with IR and TP does not characterize φ^{TTC} (see Coreno and Feng (2025, Example 1)).

6 Extension: Conditionally lexicographic preferences

Conditionally lexicographic preferences generalize purely lexicographic preferences (Booth et al. (2010); see also Domshlak et al. (2011); Pigozzi et al. (2016)). Unlike responsive preferences, they allow the relative ranking of two objects to depend on the other objects they are obtained with: for example, an agent may prefer drinking Champagne to Bordeaux when paired with oysters, but Bordeaux to Champagne otherwise. This flexibility accommodates complementarities among objects while retaining several appealing features of lexicographic preferences. In particular, conditionally lexicographic preferences admit a compact representation via *lexicographic preference trees* (LP trees), which makes them attractive from an implementation perspective.²⁷ As we shall see, PE coincides with IGE on this domain.

Loosely speaking, an agent has conditionally lexicographic preferences if, for any nonempty set of objects X disjoint from Y , there is a unique object in X which is the “lexicographically

²⁶Upper invariance requires an agent’s assignment to remain unchanged when she misrepresents only the ordering of objects ranked below her assignment at the truthful profile.

²⁷Intuitively, an LP tree is a rooted binary tree that represents conditional marginal preferences in a graphical manner. For any bundle Y , there is a unique root-to-leaf path that is consistent with Y . This path specifies the agent’s preference ordering over objects conditional on receiving Y . See Appendix A for details.

best" addition to Y from X .

Definition 9. A preference relation P_i on 2^O is *conditionally lexicographic* if, for all disjoint $X, Y \in 2^O$ with $X \neq \emptyset$, there is a unique object in X , denoted $x^* = \max_{P_i}(X \mid Y)$, such that

$$\text{for each } Z \subseteq X \setminus \{x^*\}, \quad (Y \cup x^*) P_i (Y \cup Z).$$

We call $x^* = \max_{P_i}(X \mid Y)$ agent i 's most-preferred object in X conditional on already having Y .

For each $i \in N$, let $\mathcal{CL}_i$ be the set of conditionally lexicographic preferences on 2^O , and write $\mathcal{CL} := \prod_{i \in N} \mathcal{CL}_i$ for the *conditionally lexicographic domain*. Every $P_i \in \mathcal{CL}_i$ is monotonic. Moreover, every lexicographic preference is conditionally lexicographic, and a preference is lexicographic if and only if it is both conditionally lexicographic and responsive. Formally, for each $i \in N$, $\mathcal{L}_i \subseteq \mathcal{CL}_i \subseteq \mathcal{M}_i$ (with both inclusions strict if $|O| \geq 3$), and $\mathcal{L}_i = \mathcal{CL}_i \cap \mathcal{R}_i$.

6.1 Properties: Conditional WELB and DSP

In this section we look beyond marginal rules and consider rules that depend on agents' full preferences over bundles. Among the properties we use below, the definitions of BAL, PE, IGE, and IR carry over from Section 2.2 without modification. By contrast, WELB and DSP, which were originally defined with respect to agents' marginal preferences, must be reformulated using the conditional marginal preferences introduced below.²⁸

Given $P_i \in \mathcal{CL}_i$ and a bundle $Y \subseteq O$, we define the *conditional marginal preference* $P_i(Y)|_O$ over individual objects *conditional on receiving* Y . Formally, $P_i(Y)|_O$ is the strict linear order on O such that, for all $x, y \in O$,²⁹

$$x P_i(Y)|_O y \iff (Y \cup \{x\}) \setminus \{y\} P_i (Y \cup \{y\}) \setminus \{x\}.$$

Equivalently, when P_i is represented by an LP tree, $P_i(Y)|_O$ is the order in which the objects are encountered along the unique path consistent with Y (see Appendix A). We define $R_i(Y)|_O$ by $x R_i(Y)|_O y$ if and only if $(x P_i(Y)|_O y \text{ or } x = y)$. For notational convenience, we will often write $P_i(Y)$ instead of $P_i(Y)|_O$ (and similarly $R_i(Y)$ instead of $R_i(Y)|_O$). Given a nonempty subset $X \subseteq O$, let $\max_{P_i(Y)}(X)$ and $\min_{P_i(Y)}(X)$ denote the most- and least-preferred objects in X according to $P_i(Y)|_O$, respectively. Note that when X and Y are disjoint, $\max_{P_i(Y)}(X)$ coincides with $\max_{P_i}(X \mid Y)$.

²⁸We do not discuss truncation-proofness on $\mathcal{CL}$ because truncation is no longer a simple operation on a single marginal ranking. A conditionally lexicographic preference P_i induces a family $(P_i(Y)|_O)_{Y \subseteq O}$ of conditional marginal preferences, and a truncation would have to specify compatible cutoffs across all conditioning bundles.

²⁹Equivalently, for all $x, y \in O$, $x P_i(Y)|_O y \iff x \cup (Y \setminus \{x, y\}) P_i y \cup (Y \setminus \{x, y\})$.

Remark 1. Each conditionally lexicographic P_i induces a family $(P_i(Y)|_O)_{Y \subseteq O}$ of conditional marginal preferences. However, not every family of marginal preferences arises from a conditionally lexicographic preference: Definition 9 implies that the family $(P_i(Y)|_O)_{Y \subseteq O}$ must satisfy certain consistency constraints. For example, any two conditional marginal preferences $P_i(X)|_O$ and $P_i(Y)|_O$ must top-rank the same object. The LP-tree representation in Appendix A spells out these constraints precisely. $\diamond$

An allocation μ satisfies the *worst-endowment lower bound* at a preference profile $P \in \mathcal{CL}$ if, for each $i \in N$, and each $o \in \mu_i$, $o R_i(\mu_i) \min_{P_i(\mu_i)}(\omega_i)$. In other words, *conditional on receiving μ_i* , every object assigned to agent i is at least as good, according to $P_i(\mu_i)|_O$, as her worst endowed object.³⁰ See Appendix A (especially Figures 3 and 4) for further intuition and examples that illustrate specific assignments ruled out by WELB.

Definition 10. A rule φ satisfies the *worst-endowment lower bound (WELB)* if, for each $P \in \mathcal{CL}$, $\varphi(P)$ satisfies the worst-endowment lower bound at P .

Drop strategies extend naturally to conditionally lexicographic preferences. An agent implements a drop strategy by dropping an object she does not own to the bottom of her conditionally lexicographic preferences. Formally, given $P_i \in \mathcal{CL}_i$, we say that $P'_i \in \mathcal{CL}_i$ is a *drop strategy* for P_i if there exists $x \in O \setminus \omega_i$ such that

- (i) for each nonempty $Y \subseteq O \setminus \{x\}$, $Y P'_i x$, and
- (ii) for all $Y, Z \subseteq O \setminus \{x\}$, $Y P'_i Z$ if and only if $Y P_i Z$.

In this case, we say that P'_i is obtained from P_i by dropping object x .³¹ Note that this definition is equivalent to the one in Section 2.2 when P_i is purely lexicographic. Let $\mathcal{D}_i(P_i)$ denote the set of all drop strategies for P_i .

Intuitively, if P'_i is obtained from P_i by dropping object x , then, for each $Y \subseteq O$, the conditional marginal preference $P'_i(Y)|_O$ is obtained from $P_i(Y \setminus \{x\})|_O$ by dropping object x . In particular, x is the worst object in every conditional marginal preference associated with P'_i . The LP-tree representation in Appendix A shows that this family of modified conditional marginal preferences is induced by a unique conditionally lexicographic preference $P'_i \in \mathcal{CL}_i$.

Definition 11. A rule φ satisfies *drop strategy-proofness (DSP)* if, for each $P \in \mathcal{CL}$, each $i \in N$, and each $P'_i \in \mathcal{D}_i(P_i)$, we have $\varphi_i(P) R_i \varphi_i(P'_i, P_{-i})$.

³⁰WELB and IR are logically independent on $\mathcal{CL}$. Under an auxiliary preference assumption that each agent treats bundles that fail WELB as unacceptable (i.e., strictly worse than her endowment), then IR implies WELB on the resulting domain (see footnote 13). We do not impose this additional restriction.

³¹As in the case of purely lexicographic preferences (but not responsive preferences), there is exactly one P'_i obtained from P_i by dropping x .

6.2 Augmented Top Trading Cycles

The *Augmented Top Trading Cycles rule (ATTC)*, denoted φ^{ATTC} , is the natural extension of φ^{TTC} from lexicographic to conditionally lexicographic preferences (Fujita et al. (2018)). At each step of the ATTC algorithm, each agent points to her most-preferred unassigned object *conditional on the objects already assigned to her*, and every unassigned object points to its owner. The resulting directed graph contains at least one trading cycle, and each agent involved in a trading cycle is assigned the object to which she points. All objects involved in a trading cycle are then removed. If unassigned objects remain, then the procedure continues to the next step; otherwise, it terminates with the resulting allocation.

Formally, given a preference profile $P \in \mathcal{CL}$, φ^{ATTC} returns the allocation $\varphi^{\text{ATTC}}(P)$ determined by the following *ATTC algorithm* at P . We denote this specific instance as $\text{ATTC}(P)$.

Algorithm: $\text{ATTC}(P)$

Initialization: Set $\mu^0 := (\emptyset)_{i \in N}$ and $O^1 := O$.

Step $t \geq 1$: Each agent i points to $\max_{P_i}(O^t \mid \mu_i^{t-1})$, and each object in O^t points to its owner.

Let $\mathcal{C}_t(P)$ be the resulting set of trading cycles. For each $C \in \mathcal{C}_t(P)$, assign to each $i \in N(C)$ the object she points to in C ; this yields the partial allocation μ^t with

$$\mu_i^t = \begin{cases} \mu_i^{t-1} \cup \{\max_{P_i}(O^t \mid \mu_i^{t-1})\}, & \text{if } i \in \bigcup_{C \in \mathcal{C}_t(P)} N(C), \\ \mu_i^{t-1}, & \text{otherwise.} \end{cases}$$

Remove all objects involved in trading cycles to form the set $O^{t+1} = O^t \setminus \bigcup_{C \in \mathcal{C}_t(P)} O(C)$.

Proceed to step $t + 1$ if $O^{t+1} \neq \emptyset$; proceed to Termination otherwise.

Termination: Because at least one object is removed at each step, the algorithm terminates at some step T . Return the allocation $\varphi^{\text{ATTC}}(P) := \mu^T$.

6.3 Characterization of ATTC

Although conditionally lexicographic preferences allow for complementarities, the domain still behaves like the lexicographic domain from the perspective of efficiency. In particular, ruling out Pareto-improving single-object exchanges is enough to guarantee full Pareto efficiency.

Proposition 3. *On the conditionally lexicographic domain, an allocation satisfies IGE if and only if it satisfies PE.*

The proof of Proposition 3 is straightforward and thus omitted. The equivalence between IGE and PE implies that any myopic procedure which, at each step, greedily executes a Pareto-

improving trading cycle whenever one exists must terminate at a PE allocation on this domain. The ATTC algorithm is one such procedure.

Beyond efficiency, [Fujita et al. \(2018\)](#) show that φ^{ATTC} is core selecting (and hence satisfies both PE and IR). They also prove that it is NP-hard for an agent to find a profitable manipulation, and that any successful manipulation yields only limited gains: no manipulator can obtain an object better than her most-preferred object under truthful reporting. Complementing these results, we establish that φ^{ATTC} satisfies DSP on the conditionally lexicographic domain.

Proposition 4. *On the conditionally lexicographic domain, φ^{ATTC} satisfies DSP.*

Proposition 4 extends a result of [Altuntaş et al. \(2023\)](#), which shows that φ^{TTC} satisfies a strong drop strategy-proofness requirement on the lexicographic domain—one that also protects against manipulations where agents drop objects they own. We work with the weaker DSP (that considers only drops outside the endowment), which suffices for our characterization result. Moreover, φ^{ATTC} fails the stronger variant on the conditionally lexicographic domain.

Theorem 6 says that φ^{ATTC} is the unique balanced rule satisfying PE, WELB, and DSP, extending Theorem 2 from the lexicographic domain to the broader conditionally lexicographic domain.

Theorem 6. *On the conditionally lexicographic domain, a balanced rule satisfies PE, WELB, and DSP if and only if it equals φ^{ATTC} .*

Table 3 collects the key properties that φ^{TTC} and φ^{ATTC} enjoy on the three domains we consider.

Table 3: Properties of TTC and ATTC on various domains

Domain (Rule)	BAL	WELB	IR	PE	IGE	TP	DSP
$\mathcal{L}$ (TTC)	✓	✓	✓	✓	✓	✓	✓
$\mathcal{R}$ (TTC)	✓	✓	✓		✓	✓	
$\mathcal{CL}$ (ATTC)	✓	✓	✓	✓	✓	—	✓

Notes: On $\mathcal{L}$ and $\mathcal{CL}$, PE and IGE coincide. We do not study TP on $\mathcal{CL}$. $\diamond$

We close this section with two maximal-domain results that identify the conditionally lexicographic domain as a natural boundary for possibility. The first boundary concerns efficiency. On any monotonic domain that strictly includes $\mathcal{CL}$, the equivalence between PE and IGE breaks down.

Proposition 5. *Within the domain of monotonic preferences, the conditionally lexicographic domain is a maximal domain on which IGE and PE are equivalent.*

(More precisely, if $\mathcal{CL} \subsetneq \mathcal{P} \subseteq \mathcal{M}$, then there is a set N of agents, a preference profile $P \in \mathcal{P}$, and an allocation $\mu \in \mathcal{A}$ such that μ satisfies IGE but not PE at P .)

The second boundary concerns existence. Although φ^{ATTC} is balanced and core selecting on $\mathcal{CL}$, no rule can deliver these properties on any larger domain. On any monotonic domain that strictly includes $\mathcal{CL}$, no balanced rule satisfies even PE and IR.

Proposition 6. *Within the domain of monotonic preferences, the conditionally lexicographic domain is a maximal domain on which balancedness is compatible with PE and IR.*

(More precisely, if $\mathcal{CL} \subsetneq \mathcal{P} \subseteq \mathcal{M}$, then there is a set N of agents, an endowment $\omega \in \mathcal{A}$, and a preference profile $P \in \mathcal{P}$ such that no allocation $\mu \in \mathcal{A}^{\text{BAL}}$ is both Pareto efficient and individually rational at P .)

Remark 2. The restriction to monotonic preferences in Propositions 5 and 6 is tied to our maintained convention that every agent owns and receives a nonempty bundle. If the model is extended to allow for empty endowments and empty assignments, both maximal-domain results extend to all strict preferences. In particular, $\mathcal{CL}$ is then a maximal domain (among all strict preferences) for the equivalence between PE and IGE, and also for the compatibility of BAL, PE, and IR. $\diamond$

7 Related literature and discussion

Since its introduction for housing markets in the seminal paper by Shapley and Scarf (1974), TTC has become the subject of a large body of literature; see Morrill and Roth (2024) for a survey. For housing markets with strict preferences, TTC is the canonical rule: it selects the unique core allocation (Shapley and Scarf (1974); Roth and Postlewaite (1977)), it is group strategy-proof (Roth (1982); Bird (1984)), and it is the unique rule satisfying PE, IR, and SP (Ma (1994)). Theorem 5 strengthens Ma’s (1994) characterization by replacing SP with the weaker incentive requirement TP, leaving little scope for alternative rules in housing markets; see Section 5 for further discussion.

Early work on multi-object reallocation consists mostly of negative results. On general preference domains, PE, IR, and SP are incompatible once some agent owns more than one object (Sönmez (1999)), and the impossibility persists even under lexicographic preferences (Todo et al. (2014)). Moreover, the core can be empty even under additive (thus responsive) preferences (Konishi et al. (2001)).

Recent studies obtain positive findings under restricted preference domains. For instance, PE, IR, and SP are compatible when agents’ bundle preferences are based on “dichotomous” (Andersson et al. (2021)) or “trichotomous” (Manjunath and Westkamp (2021, 2025)) marginal preferences over objects (see also Han et al. (2024)). In addition, under conditionally lexicographic preferences the core is guaranteed to be nonempty, and ATTC selects an allocation in the core (Fujita et al. (2018)). We further show that $\mathcal{CL}$ is essentially the broadest domain on which

this is possible: $\mathcal{CL}$ is a maximal domain on which a balanced rule can satisfy even PE and IR (Proposition 6).

In the closest paper to ours, Altuntaş et al. (2023) consider the same model but focus on the lexicographic domain. They show that TTC is characterized by PE, DSP, and a property termed the *strong endowment lower bound*, which requires each agent’s assignment to “pairwise dominate” her endowment according to her marginal preferences.³² The strong endowment lower bound implies BAL and WELB on any preference domain, and under responsive preferences it also implies IR. Consequently, our lexicographic-domain results identify TTC under weaker criteria: we replace their strong endowment lower bound with BAL and WELB (Theorems 1 and 2), and replace DSP with TP in Theorem 1. We also address open questions by providing characterizations on two richer domains: the responsive domain (Theorems 3 and 4) and the conditionally lexicographic domain (Theorem 6).³³

Other studies consider adaptations of TTC to multi-object reallocation environments with additional structure. Biró et al. (2022a,b) consider a multiple-copy Shapley-Scarf model, where each agent owns multiple copies of a homogeneous, agent-specific object (equivalently, every agent is indifferent between all objects with the same initial owner). Focusing on marginal rules for responsive preferences, Biró et al. (2022a) characterize a variant of TTC using subset-drop strategy-proofness, and they determine the capacity configurations under which this variant satisfies PE, IR, and SP. Similarly, Feng et al. (2024) characterize another variant of TTC for Moulin’s (1995) multiple-type housing markets, where objects are partitioned into types and each agent consumes exactly one object of each type.

Our characterizations of TTC under responsive preferences complement characterizations of strategy-proof alternatives on the responsive domain. *Sequential Dictatorships* are characterized by SP, PE, and “non-bossiness” (e.g., Pápai (2001); Ehlers and Klaus (2003); Hatfield (2009); Monte and Tumennasan (2015)), whereas *Segmented Trading Cycles rules* are characterized by SP, “strong individual rationality”, non-bossiness, and “trade sovereignty” (Pápai (2003, 2007)). Our characterizations identify TTC as a sensible middle ground: though it fails two of the three ideals on $\mathcal{R}$ (PE and SP), it delivers meaningful guarantees along all three dimensions (IGE, IR/WELB, and TP).

We relax SP by ruling out restricted classes of simple manipulations: subset-drop, drop, and truncation strategies. This focus is consistent with computational results emphasizing the difficulty of manipulating TTC in practice (e.g., Fujita et al. (2018); Phan and Purcell (2022)). Drop strategies and their variants are studied by Altuntaş et al. (2023) for multi-object

³²Formally, an allocation μ satisfies the *strong endowment lower bound* at P if, for each agent i , there is a bijection $f_i : \omega_i \rightarrow \mu_i$ such that, for each $o \in \omega_i$, $f_i(o) R_i o$.

³³Specifically, Altuntaş et al. (2023, p. 167) asked for an extension of TTC beyond the purely lexicographic domain that continues to satisfy PE, IR, and robustness against some form of manipulation, and Fujita et al. (2018, p. 531) asked for a characterization of ATTC.

reallocation under lexicographic preferences. Truncation has roots in the matching literature, where it corresponds to shortening a list of acceptable partners (Mongell and Roth (1991); Roth and Vande Vate (1991); Roth and Rothblum (1999); Ehlers (2008)). Truncation strategies also appear in matching with contracts (e.g., Afacan (2016); Hatfield et al. (2021)) and school-choice problems (Shirakawa (2025)), though our notion is closest to formulations studied by Kojima (2013) for multi-object assignment and Biró et al. (2022a,b) for multi-object reallocation.

Finally, when relaxing marginality and considering more complex reporting languages, competitive-equilibrium approaches become natural benchmarks. Existence issues are typically avoided either by seeking equilibria in pseudomarkets for probability shares (e.g., Hylland and Zeckhauser (1979); Echenique et al. (2021, 2023)) or by accepting only approximate market clearing (e.g., Budish (2011); Nguyen and Vohra (2024); Jantschgi et al. (2025)). These procedures offer compelling efficiency and fairness properties on general preference domains, but they typically suffer from onerous reporting requirements and considerable computational complexity. Our study of ATTC under conditionally lexicographic preferences provides a practical alternative: its reporting language comprises LP-trees, which are expressive but compact, and it delivers strong welfare guarantees using a computationally tractable procedure.

A Lexicographic preference trees

This appendix introduces lexicographic preference trees, which provide a compact representation of conditionally lexicographic preferences.

Definition 12. A *lexicographic preference tree* (LP tree) on a nonempty subset $X \subseteq O$ is a directed binary tree τ_i such that

1. each vertex v is labeled with an object $o(v) \in X$.
2. every object in X appears exactly once on any path from the root to a leaf.
3. every internal (non-leaf) vertex v has two outgoing edges: (i) an “in edge” labeled $o(v)$, and (ii) a “not-in edge” labeled $\neg o(v)$.

An LP tree on O is referred to more succinctly as an *LP tree*.

Given an LP tree τ_i and a bundle $X \subseteq O$, let $\tau_i(X) = (v_1, v_2, \dots, v_{|O|})$ be the unique root-to-leaf path in τ_i containing only edges consistent with X : for each $k = 1, \dots, |O| - 1$, traverse the “in edge” from v_k if $o(v_k) \in X$, and the “not-in edge” from v_k if $o(v_k) \notin X$. Given bundles A and B with $A \neq B$, let $\tau_i(A, B)$ denote the last vertex common to both $\tau_i(A)$ and $\tau_i(B)$. Equivalently, $\tau_i(A, B)$ is the first vertex v on $\tau_i(A)$ and $\tau_i(B)$ with $o(v) \in (A \setminus B) \cup (B \setminus A)$. Figures 1 and 2 provide a graphical representation of two LP trees on $O = \{a, b, c, d\}$, together with the paths consistent with $\{a, b\}$ and $\{a, c\}$.

Definition 13. The preference relation P_{τ_i} associated with an LP tree τ_i on O is defined by:

1. for all $A, B \subseteq O$ with $A \neq B$, $[A P_{\tau_i} B \iff o(\tau_i(A, B)) \in A \setminus B]$.
2. for all $A, B \subseteq O$, $[A R_{\tau_i} B \iff (A = B \text{ or } A P_{\tau_i} B)]$.

A preference relation P_i on 2^O is conditionally lexicographic if and only if there exists an LP tree τ_i (on O) such that $P_i = P_{\tau_i}$. In fact, there is a one-to-one correspondence between LP trees and conditionally lexicographic preferences on O , and we denote the unique LP tree associated with P_i by τ_{P_i} . A lexicographic preference corresponds to an LP tree where all root-to-leaf paths use the same object order (i.e., all vertices at the same depth are labeled with the same object).

Worst-endowment lower bound. Given an LP tree τ_i and a vertex v of τ_{P_i} , let $a(v)$ denote the set of objects labeling the ancestors of v , including v itself. That is, if $(v_1, v_2, \dots, v_k = v)$ is the unique path from the root of τ_i to v , then $a(v) = \{o(v_1), o(v_2), \dots, o(v_k)\}$. Given a preference relation $P_i \in \mathcal{CL}$ and a bundle $X \in 2^O$, define $w_{P_i}(\omega_i \mid X)$ as the last vertex on the path $\tau_{P_i}(X)$ whose label belongs to ω_i . Equivalently, $w_{P_i}(\omega_i \mid X)$ is the unique vertex on $\tau_{P_i}(X)$ such that $o(w_{P_i}(\omega_i \mid X)) \in \omega_i$ and $\omega_i \subseteq a(w_{P_i}(\omega_i \mid X))$. We illustrate this construction graphically in Figures 3 and 4.

An allocation μ satisfies the *worst-endowment lower bound* at a preference profile $P \in \mathcal{CL}$ if, for each $i \in N$, $\mu_i \subseteq a(w_{P_i}(\omega_i \mid \mu_i))$. This means every object in μ_i labels a vertex on $\tau_{P_i}(\mu_i)$

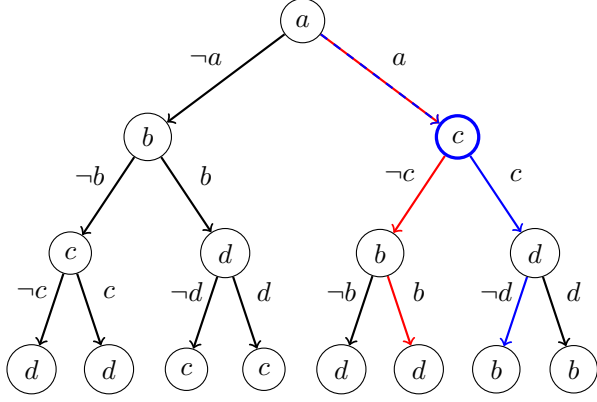


Figure 1: An LP tree τ_i

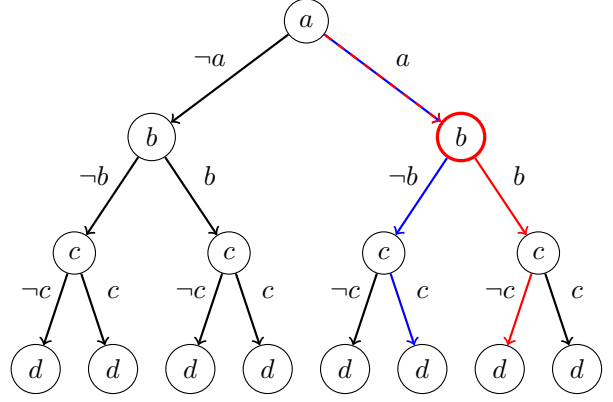


Figure 2: An LP tree τ_i^*

Notes: In both figures, the paths corresponding to the bundles $\{a, b\}$ and $\{a, c\}$ are highlighted in red and blue, respectively. In Figure 1, the last common vertex, $\tau_i(\{a, b\}, \{a, c\})$, is highlighted in blue. Because $o(\tau_i(\{a, b\}, \{a, c\})) = c$ belongs to $\{a, c\} \setminus \{a, b\}$, we have $\{a, c\} P_{\tau_i} \{a, b\}$. In Figure 2, the last common vertex, $\tau_i^*(\{a, b\}, \{a, c\})$, is highlighted in red. Because $o(\tau_i^*(\{a, b\}, \{a, c\})) = b$ belongs to $\{a, b\} \setminus \{a, c\}$, we have $\{a, b\} P_{\tau_i^*} \{a, c\}$. Since all paths in τ_i^* use the same object order, $P_{\tau_i^*}$ is purely lexicographic. $\diamond$

that appears before (or equals) vertex $w_{P_i}(\omega_i \mid \mu_i)$. Intuitively, conditional on receiving μ_i , no object assigned to agent i is worse than every object in her endowment ω_i . When agents have purely lexicographic preferences, this definition aligns with the original WELB.

Drop strategies. Given $P_i \in \mathcal{CL}_i$, let P'_i be obtained from P_i by dropping object x . Then P'_i is represented by the LP tree $\tau_{P'_i}$, obtained from τ_{P_i} by moving x to the bottom of every root-to-leaf path. Formally, let $\tau_{P_i}^{-x}$ denote the LP tree on $O \setminus \{x\}$ representing the restriction of P_i to subsets of $O \setminus \{x\}$.³⁴ To construct $\tau_{P'_i}$ from $\tau_{P_i}^{-x}$, at each leaf of $\tau_{P_i}^{-x}$ append two child nodes labeled with x , connected by an “in edge” and a “not-in edge” accordingly. This modification ensures that x is the least desirable addition to any bundle. Figures 5 and 6 illustrate this construction.

³⁴We can construct $\tau_{P_i}^{-x}$ as follows. For each vertex v of τ_{P_i} , let T_v denote the maximal subtree of τ_{P_i} consisting of a vertex v of τ_{P_i} together with all of its successors, and let v' be the child of v whose incoming edge (v, v') is labeled with $\neg o(v)$. For each vertex v of τ_{P_i} such that $o(v) = x$, simply replace T_v with the subtree $T_{v'}$ (or the empty tree if v is a leaf of τ_{P_i}).

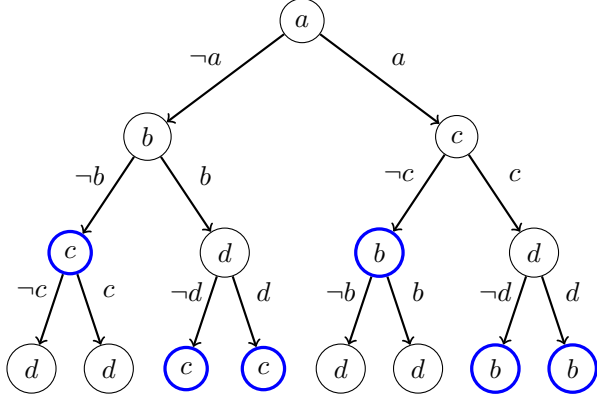


Figure 3: The LP tree τ_i

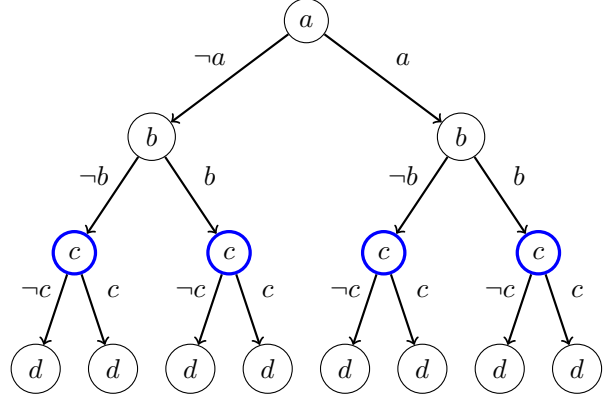


Figure 4: The LP tree τ_i^*

Notes: Figures 3 and 4 illustrate the construction of $w_{P_i}(\omega_i | X)$. Suppose agent i 's endowment is $\omega_i = \{b, c\}$. In both figures, the vertices corresponding to $w_{P_i}(\omega_i | X)$ for various $X \in 2^O$ are highlighted in blue. Notice that on any path from the root to a leaf, there is exactly one such vertex. Under τ_i , WELB precludes assigning agent i object d together with any of the bundles $\emptyset, \{a\}, \{a, b\}$, or $\{c\}$; that is, she does not receive a bundle in $\{\{d\}, \{a, d\}, \{a, b, d\}, \{c, d\}\}$. Under τ_i^* , WELB precludes any bundle containing object d . $\diamond$

B Proofs

B.1 Proof of Lemma 1

Suppose φ is a marginal rule on $\mathcal{R}$ satisfying IR. Let $P \in \mathcal{R}$ and denote $\mu := \varphi(P)$.

Toward contradiction, suppose that μ violates BAL at P . Then there exists an agent i with $|\mu_i| > |\omega_i|$. Because responsiveness imposes no restrictions on comparisons among bundles with different cardinalities (and it does not require monotonicity), there exists $P_i^* \in \mathcal{R}_i$ with $P_i^*|_O = P_i|_O$ such that, for each bundle $X \subseteq \omega_i$, $X \succ_{P_i^*} \mu_i$.³⁵ Then marginality of φ implies that $\varphi(P_i^*, P_{-i}) = \varphi(P) = \mu$. By construction of P_i^* , we have $\omega_i \succ_{P_i^*} \mu_i = \varphi_i(P_i^*, P_{-i})$, which contradicts IR. Thus φ satisfies BAL.

Now suppose for a contradiction that μ violates WELB at P . Then there is some agent $i \in N$ such that $\min_{P_i}(\omega_i) \prec_{P_i} \min_{P_i}(\mu_i)$. By BAL, we also have $|\mu_i| = |\omega_i|$. Consequently, there exists $P'_i \in \mathcal{R}_i$ with $P'_i|_O = P_i|_O$ and $\omega_i \succ_{P'_i} \mu_i$. However, marginality of φ implies that $\varphi_i(P'_i, P_{-i}) = \varphi_i(P) = \mu_i$. This contradicts IR. $\blacksquare$

B.2 Proof of Lemma 2

Suppose φ satisfies DSP and WELB. Let P'_i be obtained from P_i by dropping some subset $X \subseteq O \setminus \omega_i$. Suppose that $X = \{x_1, x_2, \dots, x_k\}$, where $x_1 \prec_{P_i} x_2 \prec_{P_i} \dots \prec_{P_i} x_k$. Then P'_i is obtained

³⁵For instance, let P_i^* order bundles first by cardinality (with smaller bundles preferred to larger bundles), and then order bundles with the same cardinality using the lexicographic order associated with $P_i|_O$. For example, if $O = \{a, b, c\}$ and $P_i|_O : a, b, c$, then $P_i^* : \emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\}$.

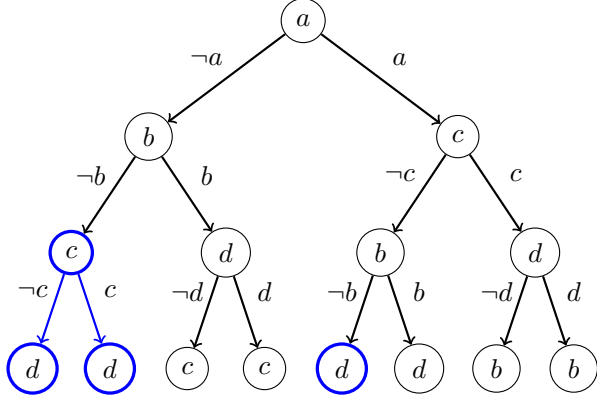


Figure 5: The LP tree τ_i

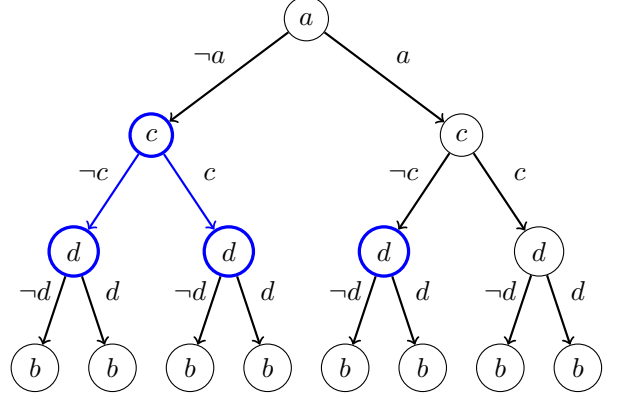


Figure 6: The LP tree τ'_i obtained by dropping b

Notes: Figure 5 displays the original LP tree τ_i from Figures 1 and 3. Suppose here that $b \notin \omega_i$ and agent i drops object b . The associated LP tree τ'_i , illustrated in Figure 6, is constructed as follows. At each internal vertex v labeled b , we do the following: (i) remove the subtree consisting of v and all of its descendants, and (ii) append the subtree, shown in blue, consisting of the child of v reached via the “not-in edge” together with all of its descendants. Then, at each leaf of this modified tree, we append two child nodes labeled b , connected via an “in edge” and a “not-in edge” accordingly. $\diamond$

from P_i via a sequence of k drop strategies. That is, $P'_i = P_i^k$, where $P_i^0 = P_i$ and $P_i^1, \dots, P_i^k$ are such that, for each $\ell \in \{1, \dots, k\}$, P_i^ℓ is obtained from $P_i^{\ell-1}$ by dropping object x_ℓ .

Claim 1. For each $\ell = 1, \dots, k$, $\varphi_i(P_i^{\ell-1}, P_{-i}) R_i \varphi_i(P_i^\ell, P_{-i})$.

Proof. The proof is by induction on ℓ . Clearly, $\varphi_i(P) = \varphi_i(P_i^0, P_{-i}) R_i \varphi_i(P_i^1, P_{-i})$ by DSP. For the inductive step, suppose that $\ell \in \{1, \dots, k-1\}$ is such that

$$\varphi_i(P) = \varphi_i(P_i^0, P_{-i}) R_i \varphi_i(P_i^1, P_{-i}) R_i \cdots R_i \varphi_i(P_i^\ell, P_{-i}).$$

It suffices to show that $\varphi_i(P_i^\ell, P_{-i}) R_i \varphi_i(P_i^{\ell+1}, P_{-i})$. By DSP, we have $\varphi_i(P_i^\ell, P_{-i}) R_i \varphi_i(P_i^{\ell+1}, P_{-i})$. Moreover, WELB implies that

$$\varphi_i(P_i^\ell, P_{-i}) \subseteq O \setminus \{x_1, \dots, x_\ell\} \quad \text{and} \quad \varphi_i(P_i^{\ell+1}, P_{-i}) \subseteq O \setminus \{x_1, \dots, x_{\ell+1}\} \subseteq O \setminus \{x_1, \dots, x_\ell\}.$$

Because $P_i^\ell|_{O \setminus \{x_1, \dots, x_\ell\}} = P_i|_{O \setminus \{x_1, \dots, x_\ell\}}$, and P_i and P_i^ℓ are lexicographic, we have $\varphi_i(P_i^\ell, P_{-i}) R_i \varphi_i(P_i^{\ell+1}, P_{-i})$, as desired. $\square$

It follows from Claim 1 that

$$\varphi_i(P) = \varphi_i(P_i^0, P_{-i}) R_i \varphi_i(P_i^1, P_{-i}) R_i \cdots R_i \varphi_i(P_i^k, P_{-i}) = \varphi_i(P'_i, P_{-i}).$$

Therefore, $\varphi_i(P) R_i \varphi_i(P'_i, P_{-i})$. $\blacksquare$

B.3 Proof of Lemma 3

Suppose φ is a marginal rule on $\mathcal{R}$ satisfying WELB and TP. Let $P \in \mathcal{R}$ and denote $\mu := \varphi(P)$.

Toward contradiction, suppose that $\mu := \varphi(P)$ violates BAL at P . Then there is an agent $i \in N$ such that $|\mu_i| > |\omega_i|$. Because responsiveness imposes no restrictions on comparisons among bundles with different cardinalities (and it does not require monotonicity), there exists $P_i^* \in \mathcal{R}_i$ such that $P_i^*|_O = P_i|_O$ and, for each bundle $X \subseteq \omega_i$, $X P_i^* \mu_i$ (see footnote 35). Then marginality of φ implies that $\varphi(P_i^*, P_{-i}) = \varphi(P) = \mu$.

Now let $P'_i \in \mathcal{R}_i$ be a truncation of P_i^* obtained by dropping the set $O \setminus \omega_i$ (thus $P'_i|_O$ ranks every object in ω_i above all others). Then WELB implies that $\varphi_i(P'_i, P_{-i}) \subseteq \omega_i$. By construction of P_i^* , we have $\varphi_i(P'_i, P_{-i}) P_i^* \mu_i = \varphi_i(P_i^*, P_{-i})$. Thus φ fails TP, a contradiction. $\blacksquare$

B.4 Proof of Theorem 1

Toward contradiction, suppose that φ satisfies the properties but $\varphi \neq \varphi^{\text{TTC}}$. We will select a preference profile according to a particular minimality criterion, and then show that the minimality of the profile leads to a contradiction. We start by introducing some notation.

For each $P \in \mathcal{L}$ and each $t \in \mathbb{N}$, recall that $\mathcal{C}_t(P)$ denotes the set of trading cycles that arise at step t of $\text{TTC}(P)$. (We assume that $\mathcal{C}_t(P) = \emptyset$ if $\text{TTC}(P)$ terminates before step t .) Define the “size” function $s : \mathcal{L} \rightarrow \mathbb{N}$ such that, for each $P \in \mathcal{L}$,

$$s(P) = \sum_{i \in N} \left| \left\{ o \in O \mid o R_i \min_{P_i}(\omega_i) \right\} \right|.$$

The “similarity” function $\rho : \mathcal{L} \rightarrow \mathbb{N} \cup \{\infty\}$ measures the earliest point of divergence between φ and $\text{TTC}(P)$. Formally, for each $P \in \mathcal{L}$,

$$\rho(P) = \inf\{t \in \mathbb{N} \mid \varphi(P) \text{ does not execute each trading cycle in } \mathcal{C}_t(P)\},$$

where it is understood that $\rho(P) = \infty$ precisely when $\varphi(P) = \varphi^{\text{TTC}}(P)$.

Among all profiles that minimize the similarity function ρ , let P be one that further minimizes the size function s . Thus, for any profile $P' \in \mathcal{L}$, either (i) $\rho(P) < \rho(P')$, or (ii) $\rho(P) = \rho(P')$ and $s(P) \leq s(P')$.

Let $t := \rho(P)$. Then $\varphi(P)$ executes each trading cycle in $\bigcup_{\tau=1}^{t-1} \mathcal{C}_\tau(P)$, but it does not execute some trading cycle $C \in \mathcal{C}_t(P)$. Suppose that

$$C = (o_1, i_1, o_2, i_2, \dots, o_k, i_k, o_{k+1} = o_1).$$

Because $\varphi(P)$ does not execute C , there is an agent $i_\ell \in N(C)$ who is not assigned the object she points to within C , i.e., $o_{\ell+1} \notin \varphi_{i_\ell}(P)$. Without loss of generality, let $i_\ell = i_k$. Thus,

$o_{k+1} = o_1 \notin \varphi_{i_k}(P)$.

Let O^t denote the set of objects available at the beginning of Step t of $\text{TTC}(P)$, i.e., $O^t := O \setminus \left(\bigcup_{\tau=1}^{t-1} \bigcup_{C' \in \mathcal{C}_\tau(P)} O(C') \right)$. Then $\varphi_{i_k}^{\text{TTC}}(P) \setminus O^t$ is the set of objects assigned to agent i_k before step t of $\text{TTC}(P)$; moreover $\varphi_{i_k}(P) \setminus O^t = \varphi_{i_k}^{\text{TTC}}(P) \setminus O^t$.

Claim 2. $\varphi_{i_k}(P) \cap O^t = \omega_{i_k} \cap O^t$.

Proof. Suppose otherwise. By BAL and the fact that $|\varphi_{i_k}(P) \setminus O^t| = |\omega_{i_k} \setminus O^t|$, we must have $|\varphi_{i_k}(P) \cap O^t| = |\omega_{i_k} \cap O^t|$. Consequently, $\varphi_{i_k}(P) \cap O^t \neq \omega_{i_k} \cap O^t$ implies there is an object $o' \in \varphi_{i_k}(P) \cap O^t$ with $o' \notin \omega_{i_k}$.

By the definition of $\text{TTC}(P)$, we know that $o_1 = \max_{P_{i_k}}(O^t)$. Because $o' \in \varphi_{i_k}(P) \cap O^t$ and $o_1 \notin \varphi_{i_k}(P)$, it follows that $o_1 P_{i_k} o'$. Furthermore, WELB implies that $o' P_{i_k} \min_{P_{i_k}}(\omega_{i_k})$.

Let P'_{i_k} be the truncation of P_{i_k} at o_1 . Thus, P'_{i_k} is obtained from P_{i_k} by dropping the set $\{o \in O \setminus \omega_{i_k} \mid o_1 P_{i_k} o\}$.

Let $P' := (P'_{i_k}, P_{-i_k})$. Then $o_1 P_{i_k} o' P_{i_k} \min_{P_{i_k}}(\omega_{i_k})$ implies that $s(P') < s(P)$. Consequently, the choice of P implies that $\rho(P') > \rho(P) = t$.

By definition of P' , the two runs $\text{TTC}(P)$ and $\text{TTC}(P')$ execute precisely the same trading cycles at every step $\tau \leq t$, i.e., $\mathcal{C}_\tau(P) = \mathcal{C}_\tau(P')$ for $\tau = 1, \dots, t$. Thus, $\rho(P') > t$ implies that $\varphi(P')$ executes all trading cycles in $\bigcup_{\tau=1}^t \mathcal{C}_\tau(P)$. In particular,

$$(\varphi_{i_k}(P) \setminus O^t) \cup \{o_1\} \subseteq \varphi_{i_k}(P').$$

Since P_{i_k} is lexicographic and $\max_{P_{i_k}}(O^t) = o_1 \notin \varphi_{i_k}(P)$, we must have $\varphi_{i_k}(P') P_{i_k} \varphi_{i_k}(P)$. This contradicts TP. $\square$

Next, we show that at least two agents are involved in C .

Claim 3. $|N(C)| > 1$.

Proof. Toward contradiction, suppose that $|N(C)| = 1$. Thus $i_1 = i_k$ and $C = (o_1, i_1, o_1)$. Then $o_1 \in \omega_{i_k}$ and Claim 2 implies that $o_1 \in \varphi_{i_k}(P)$, a contradiction. $\square$

By Claim 2 and Claim 3, agent i_{k-1} points to o_k on C , yet she does not receive o_k . An argument similar to the proof of Claim 2 implies that $\varphi_{i_{k-1}}(P) \cap O^t = \omega_{i_{k-1}} \cap O^t$.

Proceeding by induction, we conclude that for each agent i_ℓ involved in C , we have $\varphi_{i_\ell}(P) \cap O^t = \omega_{i_\ell} \cap O^t$. Thus, C is a Pareto-improving trading cycle at P . It follows that φ is not IGE, a contradiction. $\blacksquare$

B.5 Proof of Proposition 1

Fix $P \in \mathcal{R}$ and $i \in N$, and let P'_i be a truncation of P_i at an object $x \in O$.³⁶ Denote $P' := (P'_i, P_{-i})$ and $U := \{o \in O \mid o R_i x\}$, and observe that $P_i|_U = P'_i|_U$. Consider two cases.

- Case 1. Suppose $\varphi_i^{\text{TTC}}(P) \subseteq U$. Then, by the definition of $\text{TTC}(P)$ and $\text{TTC}(P')$, the allocations $\varphi^{\text{TTC}}(P)$ and $\varphi^{\text{TTC}}(P')$ execute precisely the same trading cycles. Thus $\varphi_i^{\text{TTC}}(P) = \varphi_i^{\text{TTC}}(P')$.
- Case 2. Suppose $\varphi_i^{\text{TTC}}(P) \not\subseteq U$. Then agent i receives at least one object in $O \setminus U$ at $\varphi^{\text{TTC}}(P)$. Let t be the earliest step of $\text{TTC}(P)$ at which agent i points to an object in $O \setminus U$. Let O^t be the set of objects remaining at the beginning of step t of $\text{TTC}(P)$. Since $P_i|_U = P'_i|_U$, we know that $\text{TTC}(P)$ and $\text{TTC}(P')$ execute the same trading cycles at steps $1, \dots, t-1$. Thus, O^t is the set of objects remaining at step t of $\text{TTC}(P')$. Moreover, we have

$$\varphi_i^{\text{TTC}}(P) \setminus O^t = \varphi_i^{\text{TTC}}(P') \setminus O^t. \quad (1)$$

By construction, P'_i ranks every object in $\omega_i \cap O^t$ above every object in $O^t \setminus \omega_i$. Thus, at each step $\tau \geq t$ of $\text{TTC}(P')$ where agent i is involved in a trading cycle, she is assigned an object in ω_i . Consequently,

$$\varphi_i^{\text{TTC}}(P') \cap O^t = \omega_i \cap O^t. \quad (2)$$

On the other hand, at each step $\tau \geq t$ of $\text{TTC}(P)$ where agent i is involved in a trading cycle, she is assigned an object weakly better than the object she relinquishes from her endowment. In other words, there is a bijection $\sigma : \omega_i \cap O^t \rightarrow \varphi_i^{\text{TTC}}(P) \cap O^t$ such that, for each $o \in \omega_i \cap O^t$, $\sigma(o) R_i o$.

Together with (1) and (2), we conclude that there is a bijection $\pi : \varphi_i^{\text{TTC}}(P') \rightarrow \varphi_i^{\text{TTC}}(P)$ such that, for each $o \in \varphi_i^{\text{TTC}}(P')$, $\pi(o) R_i o$. Since P_i is responsive, we must have $\varphi_i^{\text{TTC}}(P) R_i \varphi_i^{\text{TTC}}(P')$. ■

B.6 Proof of Proposition 2

Let $P \in \mathcal{R}$, $i \in N$, and $P'_i \in \mathcal{R}_i$ be such that $\varphi_i^{\text{TTC}}(P'_i, P_{-i}) P_i \varphi_i^{\text{TTC}}(P)$.

- (i) Let $P_i|_O : o_1, \dots, o_{|\omega_i|}, \dots$. In particular, $P_i|_O$ ranks every object in $\{o_1, \dots, o_{|\omega_i|}\}$ above every object in $O \setminus \{o_1, \dots, o_{|\omega_i|}\}$. For each $j \in N \setminus \{i\}$, let $P_j^*|_O : \dots, o_1, \dots, o_{|\omega_i|}$. In particular, $P_j^*|_O$ ranks every object in $O \setminus \{o_1, \dots, o_{|\omega_i|}\}$ above every object in $\{o_1, \dots, o_{|\omega_i|}\}$. Then $\varphi_i^{\text{TTC}}(P_i, P_{-i}^*) = \{o_1, \dots, o_{|\omega_i|}\}$ (otherwise φ^{TTC} would not satisfy IGE). Since P_i is responsive, $\{o_1, \dots, o_{|\omega_i|}\}$ is the most-preferred $|\omega_i|$ -subset of O according to P_i . Consequently, $B_{P_i}(\mathcal{O}_i(P_i)) R_i B_{P_i}(\mathcal{O}_i(P'_i))$.

³⁶If P'_i is obtained from P_i by dropping $O \setminus \omega_i$, then trivially $\varphi_i^{\text{TTC}}(P) R_i \varphi_i^{\text{TTC}}(P'_i, P_{-i}) = \omega_i$.

- (ii) For each $j \in N \setminus \{i\}$, let $\omega_j = \{o_1^j, \dots, o_{|\omega_j|}^j\}$ and $P_j^*|_O : o_1^j, \dots, o_{|\omega_j|}^j, \dots$. In particular, $P_j^*|_O$ ranks every object in ω_j above every object in $O \setminus \omega_j$. Then $\varphi_i^{\text{TTC}}(P_i, P_{-i}^*) = \varphi_i^{\text{TTC}}(P_i', P_{-i}^*) = \omega_i$. Consequently, since φ^{TTC} satisfies IR on $\mathcal{R}$, we have $W_{P_i}(\mathcal{O}_i(P_i)) = \omega_i R_i W_{P_i}(\mathcal{O}_i(P_i'))$.

■

B.7 Proof of Proposition 4

Fix an agent $i \in N$. To prove that i cannot manipulate using drop strategies, we consider an alternative procedure that proceeds exactly as in the ATTC algorithm except that, at each step in which multiple trading cycles arise, it executes only the cycles that do not involve i . Thus any trading cycle involving agent i is temporarily deferred, and it is executed only when it is the unique remaining cycle. It is easy to see that the order in which cycles are executed does not affect the final allocation (as any deferred trading cycle persists until it is executed). Hence this modified procedure yields the same allocation as the original ATTC algorithm.³⁷

Formally, given a preference profile $P \in \mathcal{CL}$, the allocation $\varphi^{\text{ATTC}}(P)$ is equivalently obtained via the following algorithm, which we denote $\text{ATTC}^i(P)$. The algorithm defines, for each step t , several ingredients that will be used throughout the proof: (i) the remaining set of objects $O^t(P)$, (ii) the directed graph $G^t(P)$, (iii) the full set $\tilde{\mathcal{C}}_t(P)$ of cycles that *arise* at step t , (iv) the subset $\bar{\mathcal{C}}_t(P)$ of trading cycles that are *executed* at step t , and (v) the partial allocation $\mu^t(P)$ constructed by the end of step t .

Algorithm: $\text{ATTC}^i(P)$

Initialization: Set $\mu^0(P) := (\emptyset)_{j \in N}$ and $O^1(P) := O$.

Step $t \geq 1$: Construct a directed graph $G^t(P)$ on $N \cup O^t(P)$ as follows. Each agent j points to $\max_{P_j}(O^t(P) \mid \mu_j^{t-1}(P))$, and each object in $O^t(P)$ points to its owner. Let $\tilde{\mathcal{C}}_t(P)$ be the resulting set of trading cycles, and let

$$\bar{\mathcal{C}}_t(P) = \begin{cases} \{C \in \tilde{\mathcal{C}}_t(P) \mid i \notin N(C)\}, & \text{if } |\tilde{\mathcal{C}}_t(P)| \geq 2 \\ \tilde{\mathcal{C}}_t(P), & \text{if } |\tilde{\mathcal{C}}_t(P)| = 1. \end{cases}$$

Execute all cycles in $\bar{\mathcal{C}}_t(P)$, yielding the partial allocation $\mu^t(P)$ with

$$\mu_j^t(P) = \begin{cases} \mu_j^{t-1}(P) \cup \{\max_{P_j}(O^t(P) \mid \mu_j^{t-1}(P))\}, & \text{if } j \in \bigcup_{C \in \bar{\mathcal{C}}_t(P)} N(C) \\ \mu_j^{t-1}(P), & \text{otherwise.} \end{cases}$$

Remove all objects involved in executed trading cycles to form the set $O^{t+1}(P) = O^t(P) \setminus$

³⁷A similar modification is studied by [Gonczarowski et al. \(2023\)](#) in single-object environments.

$\cup_{C \in \bar{\mathcal{C}}_t(P)} O(C)$. Proceed to step $t + 1$ if $O^{t+1}(P) \neq \emptyset$; proceed to Termination otherwise.
Termination: Let T be the earliest step with $O^{T+1}(P) = \emptyset$. Return $\varphi^{\text{ATTC}}(P) := \mu^T(P)$.

Let $P \in \mathcal{CL}$. Let $\tilde{P}_i \in \mathcal{D}_i(P_i)$ be obtained from P_i by dropping an object $x \in O \setminus \omega_i$, and denote $\tilde{P} := (\tilde{P}_i, P_{-i})$. We prove that $\varphi_i^{\text{ATTC}}(P) R_i \varphi_i^{\text{ATTC}}(\tilde{P})$ by comparing the two runs of the modified ATTC algorithm.³⁸

If $\bar{\mathcal{C}}_t(P) = \bar{\mathcal{C}}_t(\tilde{P})$ for every step t , then the two runs execute exactly the same cycles; hence $\varphi^{\text{ATTC}}(P) = \varphi^{\text{ATTC}}(\tilde{P})$ and the claim holds.

Otherwise, let t be the earliest step such that $\bar{\mathcal{C}}_t(P) \neq \bar{\mathcal{C}}_t(\tilde{P})$. Then $\bar{\mathcal{C}}_s(P) = \bar{\mathcal{C}}_s(\tilde{P})$ for each step $s < t$, which implies that the partial allocations and the remaining objects coincide at the beginning of step t in the two runs, i.e.,

$$\mu^{t-1}(P) = \mu^{t-1}(\tilde{P}) \quad \text{and} \quad O^t(P) = O^t(\tilde{P}).$$

Consider the directed graphs $G^t(P)$ and $G^t(\tilde{P})$ constructed at step t in the two runs. Because $P_{-i} = \tilde{P}_{-i}$, $\mu^{t-1}(P) = \mu^{t-1}(\tilde{P})$, and $O^t(P) = O^t(\tilde{P})$, every agent $j \in N \setminus \{i\}$ points to the same object in both graphs. Hence, the sets of cycles not involving i coincide:

$$\{C \in \tilde{\mathcal{C}}_t(P) \mid i \notin N(C)\} = \{C \in \tilde{\mathcal{C}}_t(\tilde{P}) \mid i \notin N(C)\}. \quad (3)$$

If the common set in (3) were nonempty, then by definition of $\text{ATTC}^i(P)$ and $\text{ATTC}^i(\tilde{P})$ we would have

$$\bar{\mathcal{C}}_t(P) = \{C \in \tilde{\mathcal{C}}_t(P) \mid i \notin N(C)\} = \{C \in \tilde{\mathcal{C}}_t(\tilde{P}) \mid i \notin N(C)\} = \bar{\mathcal{C}}_t(\tilde{P}),$$

contradicting the choice of t . Thus

$$\{C \in \tilde{\mathcal{C}}_t(P) \mid i \notin N(C)\} = \{C \in \tilde{\mathcal{C}}_t(\tilde{P}) \mid i \notin N(C)\} = \emptyset.$$

Since agent i can be involved in at most one cycle in $G^t(P)$ and in $G^t(\tilde{P})$, and each graph contains a cycle, we must have $\bar{\mathcal{C}}_t(P) = \tilde{\mathcal{C}}_t(P) = \{C\}$ and $\bar{\mathcal{C}}_t(\tilde{P}) = \tilde{\mathcal{C}}_t(\tilde{P}) = \{\tilde{C}\}$ for some trading cycles $C, \tilde{C}$ with $i \in N(C) \cap N(\tilde{C})$. Moreover, $C \neq \tilde{C}$ because $\bar{\mathcal{C}}_t(P) \neq \bar{\mathcal{C}}_t(\tilde{P})$. Thus agent i must point to different objects in $G^t(P)$ and $G^t(\tilde{P})$ (since all other agents point to the same object). Because $Y := \mu_i^{t-1}(P) = \mu_i^{t-1}(\tilde{P})$ and $\tilde{P}_i$ is obtained from P_i by dropping object x , the conditional marginal preferences $P_i(Y)|_O$ and $\tilde{P}_i(Y)|_O$ agree on $O \setminus \{x\}$; hence if i points to

³⁸A direct step-by-step comparison of $\text{ATTC}(P)$ and $\text{ATTC}(\tilde{P})$ is awkward because agent i may receive the same object at different steps in the two runs, creating a timing mismatch. The modified algorithm is used to synchronize the two runs $\text{ATTC}^i(P)$ and $\text{ATTC}^i(\tilde{P})$ up to the earliest step at which i receives a different object.

different objects in the two graphs, then she necessarily points to x in C but not in $\tilde{C}$. Since C is the unique cycle executed at step t of $\text{ATTC}^i(P)$, we have $x \in \varphi_i^{\text{ATTC}}(P)$.

Because $x \notin \omega_i$ and $\tilde{P}_i$ is obtained from P_i by dropping x , the object x is ranked below $\min_{\tilde{P}_i(Z)}(\omega_i)$ according to every conditional marginal preference $\tilde{P}_i(Z)|_O$ associated with $\tilde{P}_i$. Since φ^{ATTC} satisfies WELB, we must have $x \notin \varphi_i^{\text{ATTC}}(\tilde{P})$.

Finally, since agent i points to x at step t of $\text{ATTC}^i(P)$, we have $x = \max_{P_i}(O^t(P) \mid \mu_i^{t-1}(P))$. Since P_i is conditionally lexicographic and $x \in \varphi_i^{\text{ATTC}}(P) \setminus \varphi_i^{\text{ATTC}}(\tilde{P})$, agent i strictly prefers to complete her partial assignment $\mu_i^{t-1}(P)$ by adding the set $\varphi_i^{\text{ATTC}}(P) \cap O^t(P)$ instead of $\varphi_i^{\text{ATTC}}(\tilde{P}) \cap O^t(\tilde{P})$. Since

$$\begin{aligned} \varphi_i^{\text{ATTC}}(P) &= \mu_i^{t-1}(P) \cup (\varphi_i^{\text{ATTC}}(P) \cap O^t(P)) \\ \text{and } \varphi_i^{\text{ATTC}}(\tilde{P}) &= \mu_i^{t-1}(P) \cup (\varphi_i^{\text{ATTC}}(\tilde{P}) \cap O^t(\tilde{P})), \end{aligned}$$

we conclude that $\varphi_i^{\text{ATTC}}(P) P_i \varphi_i^{\text{ATTC}}(\tilde{P})$. ■

B.8 Proof of Theorem 6

Toward contradiction, suppose that φ satisfies the properties but $\varphi \neq \varphi^{\text{ATTC}}$. The argument here mirrors the proof of Theorem 1 for the lexicographic domain (see the sketch in Section 3).

For each $P \in \mathcal{CL}$ and each $t \in \mathbb{N}$, recall that $\mathcal{C}_t(P)$ denotes the set of trading cycles that arise at step t of $\text{ATTC}(P)$. (We assume that $\mathcal{C}_t(P) = \emptyset$ if $\text{ATTC}(P)$ terminates before step t .) The “size” function $s : \mathcal{CL} \rightarrow \mathbb{N}$ is defined, for each $P \in \mathcal{CL}$, by

$$s(P) = \sum_{i \in N} \sum_{Y \in 2^O} \left| \{o \in O \mid o R_i(Y) \min_{P_i(Y)}(\omega_i)\} \right|,$$

The “similarity” function $\rho : \mathcal{CL} \rightarrow \mathbb{N} \cup \{\infty\}$ is such that, for each $P \in \mathcal{CL}$,

$$\rho(P) = \inf\{t \in \mathbb{N} \mid \varphi(P) \text{ does not execute each trading cycle in } \mathcal{C}_t(P)\},$$

where it is understood that $\rho(P) = \infty$ precisely when $\varphi(P) = \varphi^{\text{ATTC}}(P)$.

Among all profiles that minimize the similarity function ρ , let P be one that further minimizes the size function s . Thus, for any profile $P' \in \mathcal{CL}$, either (i) $\rho(P) < \rho(P')$, or (ii) $\rho(P) = \rho(P')$ and $s(P) \leq s(P')$.

Let $t := \rho(P)$. Then $\varphi(P)$ executes each trading cycle in $\bigcup_{\tau=1}^{t-1} \mathcal{C}_\tau(P)$, but it does not execute some trading cycle $C \in \mathcal{C}_t(P)$. Suppose that

$$C = (o_1, i_1, o_2, i_2, \dots, o_k, i_k, o_{k+1} = o_1).$$

Because $\varphi(P)$ does not execute C , there is an agent $i_\ell \in N(C)$ who is not assigned the object she points to within C , i.e., $o_{\ell+1} \notin \varphi_{i_\ell}(P)$. Without loss of generality, let $i_\ell = i_k$. Thus, $o_{k+1} = o_1 \notin \varphi_{i_k}(P)$.

Let O^t denote the set of objects available at the beginning of Step t of $\text{ATTC}(P)$, i.e., $O^t := O \setminus \left(\bigcup_{\tau=1}^{t-1} \bigcup_{C' \in \mathcal{C}_\tau(P)} O(C') \right)$. Then $\varphi_{i_k}^{\text{ATTC}}(P) \setminus O^t$ is the set of objects assigned to agent i_k before step t of $\text{ATTC}(P)$; moreover $\varphi_{i_k}(P) \setminus O^t = \varphi_{i_k}^{\text{ATTC}}(P) \setminus O^t$.

Claim 4. $\varphi_{i_k}(P) \cap O^t = \omega_{i_k} \cap O^t$.

Proof. Suppose otherwise. By BAL and the fact that $|\varphi_{i_k}(P) \setminus O^t| = |\omega_{i_k} \setminus O^t|$, we must have $|\varphi_{i_k}(P) \cap O^t| = |\omega_{i_k} \cap O^t|$. Consequently, $\varphi_{i_k}(P) \cap O^t \neq \omega_{i_k} \cap O^t$ implies there is an object $o' \in \varphi_{i_k}(P) \cap O^t$ with $o' \notin \omega_{i_k}$.

By the definition of $\text{ATTC}(P)$, we know that $o_1 = \max_{P_{i_k}}(O^t \mid \varphi_{i_k}(P) \setminus O^t)$. Because $o' \in \varphi_{i_k}(P) \cap O^t$ and $o_1 \notin \varphi_{i_k}(P)$, it follows that $o_1 P_{i_k}(\varphi_{i_k}(P) \setminus O^t) o'$ (thus $o_1 \neq o'$). Furthermore, WELB implies that $o' P_{i_k}(\varphi_{i_k}(P)) \min_{P_{i_k}(\varphi_{i_k}(P))}(\omega_{i_k})$.

Let P'_{i_k} be the drop strategy obtained from P_{i_k} by dropping object o' . Let $P' := (P'_{i_k}, P_{-i_k})$. Then, for each $Y \subseteq O$,

$$\left| \{o \in O \mid o R'_{i_k}(Y) \min_{P'_{i_k}(Y)}(\omega_{i_k})\} \right| \leq \left| \{o \in O \mid o R_{i_k}(Y) \min_{P_{i_k}(Y)}(\omega_{i_k})\} \right|,$$

with strict inequality for $Y = \varphi_{i_k}(P)$ because $o' P_{i_k}(\varphi_{i_k}(P)) \min_{P_{i_k}(\varphi_{i_k}(P))}(\omega_{i_k})$. Thus $s(P') < s(P)$. Consequently, the choice of P implies that $\rho(P') > \rho(P) = t$.

By definition of P'_{i_k} , the two conditional marginal preferences $P'_{i_k}(\varphi_{i_k}(P) \setminus O^t)$ and $P_{i_k}(\varphi_{i_k}(P) \setminus O^t)$ agree for all objects at least as good as o_1 . Consequently, at every step $\tau \leq t$ of the two runs $\text{ATTC}(P)$ and $\text{ATTC}(P')$, agent i_k points to the same object, and hence the two runs execute precisely the same trading cycles at every step $\tau \leq t$, i.e., $\mathcal{C}_\tau(P) = \mathcal{C}_\tau(P')$ for $\tau = 1, \dots, t$. Thus, $\rho(P') > t$ implies that $\varphi(P')$ executes all trading cycles in $\bigcup_{\tau=1}^t \mathcal{C}_\tau(P)$. In particular,

$$(\varphi_{i_k}(P) \setminus O^t) \cup \{o_1\} \subseteq \varphi_{i_k}(P').$$

Since P_{i_k} is conditionally lexicographic and $\max_{P_{i_k}}(O^t \mid \varphi_{i_k}(P) \setminus O^t) = o_1 \notin \varphi_{i_k}(P)$, Definition 9 implies that $\varphi_{i_k}(P') P_{i_k} \varphi_{i_k}(P)$. This contradicts DSP. $\square$

Next, we show that at least two agents are involved in C .

Claim 5. $|N(C)| > 1$.

Proof. Toward contradiction, suppose that $|N(C)| = 1$. Thus $i_1 = i_k$ and $C = (o_1, i_1, o_1)$. Then $o_1 \in \omega_{i_k}$ and Claim 4 implies that $o_1 \in \varphi_{i_k}(P)$, a contradiction. $\square$

By Claim 4 and Claim 5, agent i_{k-1} points to o_k on C , yet she does not receive o_k . An argument similar to the proof of Claim 4 implies that $\varphi_{i_{k-1}}(P) \cap O^t = \omega_{i_{k-1}} \cap O^t$.

Proceeding by induction, we conclude that for each agent i_ℓ involved in C , we have $\varphi_{i_\ell}(P) \cap O^t = \omega_{i_\ell} \cap O^t$. Thus, C is a Pareto-improving trading cycle at P . It follows that φ is not IGE, a contradiction. $\blacksquare$

B.9 Proof of Proposition 5

Without loss of generality, let $P_1 \in \mathcal{M}_1 \setminus \mathcal{CL}_1$. Since P_1 is not conditionally lexicographic, there exist disjoint subsets $X, Y \in 2^O$ with $X \neq \emptyset$ such that

$$\text{for all } x \in X, \text{ there exists } Z_x \subseteq X \setminus x \text{ such that } (Y \cup Z_x) P_1 (Y \cup x).$$

Because P_1 is monotonic, $Z_x \subseteq X \setminus x$ implies $[Y \cup (X \setminus x)] R_1 (Y \cup Z_x)$. Thus, we have

$$\text{for all } x \in X, \quad [Y \cup (X \setminus x)] P_1 (Y \cup x).$$

Note also that $|X| \geq 3$.³⁹

Let $x^* \in X$ be such that $(Y \cup x^*) R_1 (Y \cup x)$ for all $x \in X$. Then, since $|X| \geq 3$ and P_1 is monotonic, it follows that

$$\text{for all } x \in X, \quad [Y \cup (X \setminus x)] P_1 (Y \cup x^*) R_1 (Y \cup x).$$

Case 1. Suppose $X \cup Y = O$. Let $N = \{1, 2\}$, and let $P_2 \in \mathcal{L}$ be such that (i) $\max_{P_2}(O) = x^*$, and (ii) for all $x \in X$, $x P_2 Y$. Then, in particular,

$$x^* P_2 [Y \cup (X \setminus x^*)] R_2 (X \setminus x^*).$$

Consider the allocations

$$\mu := (Y \cup x^*, X \setminus x^*) \quad \text{and} \quad \bar{\mu} := (Y \cup (X \setminus x^*), x^*).$$

Then μ is not PE at P because it is Pareto dominated by $\bar{\mu}$ at P .

We claim that μ satisfies IGE at P . Indeed, consider any trading cycle $C = (x, 2, y, 1, x)$, where $x \in \mu_2$ and $y \in \mu_1$. Executing C yields the allocation μ' , where

$$\mu'_1 = (Y \cup \{x^*, x\}) \setminus y \quad \text{and} \quad \mu'_2 = (X \setminus \{x^*, x\}) \cup y.$$

³⁹If X is a singleton, say $X = \{x\}$, then $Y P_1 (Y \cup x)$, a violation of monotonicity. If X contains exactly two objects, say $X = \{x, y\}$, then $(Y \cup y) P_1 (Y \cup x) P_1 (Y \cup y)$, a violation of transitivity.

If $y \in Y$, then this exchange harms agent 2, i.e., $\mu_2 P_2 \mu'_2$. On the other hand, if $y = x^*$, then this exchange harms agent 1 because $\mu_1 = (Y \cup x^*) P_1 (Y \cup x) = \mu'_1$. Hence, μ satisfies IGE.

Case 2. Suppose $X \cup Y \subsetneq O$. Let $N = \{1, 2, 3\}$ and denote $\overline{O} := O \setminus (X \cup Y)$. Let $P_2 \in \mathcal{L}$ be such that (i) $\max_{P_2}(O) = x^*$, and (ii) for all $x \in X$, $x P_2 (O \setminus X)$. Let $P_3 \in \mathcal{L}$ be such that (i) for all $z \in \overline{O}$, $z P_3 (X \cup Y)$.

Consider the allocations

$$\mu := (Y \cup x^*, X \setminus x^*, \overline{O}) \quad \text{and} \quad \bar{\mu} := (Y \cup (X \setminus x^*), x^*, \overline{O}).$$

Then μ is not PE at P because it is Pareto dominated by $\bar{\mu}$ at P .

We claim that μ satisfies IGE at P . Because μ_3 is agent 3's top-ranked $|\overline{O}|$ -subset of O , any trading cycle at μ involving agent 3 must harm agent 3. Thus, any Pareto-improving trading cycle at μ must involve only agents 1 and 2. However, there is no such exchange by the argument in Case 1. Hence, μ satisfies IGE. $\blacksquare$

B.10 Proof of Proposition 6

Without loss of generality, let $P_1 \in \mathcal{M}_1 \setminus \mathcal{CL}_1$. Since P_1 is not conditionally lexicographic, there exist disjoint subsets $X, Y \in 2^O$ with $X \neq \emptyset$ such that

$$\text{for all } x \in X, \text{ there exists } Z_x \subseteq X \setminus x \text{ such that } (Y \cup Z_x) P_1 (Y \cup x).$$

Because P_1 is monotonic, $Z_x \subseteq X \setminus x$ implies $[Y \cup (X \setminus x)] R_1 (Y \cup Z_x)$. Thus, we have

$$(i) \ X \neq \emptyset = X \cap Y, \quad \text{and} \quad (ii) \ [Y \cup (X \setminus x)] P_1 (Y \cup x) \quad \text{for every } x \in X. \quad (4)$$

Among all pairs (X, Y) of bundles satisfying (4), choose (X, Y) such that $|Y|$ is minimal. Note that $|X| \geq 3$ (see footnote 39 in the proof of Proposition 5).

Let $x^* \in X$ be such that $(Y \cup x^*) R_1 (Y \cup x)$ for every $x \in X$. Then, since $|X| \geq 3$ and P_1 is monotonic, it follows that

$$\text{for all } x \in X, \quad [Y \cup (X \setminus x)] P_1 (Y \cup x^*) R_1 (Y \cup x).$$

In particular, we have

$$[Y \cup (X \setminus x^*)] P_1 (Y \cup x^*) R_1 (Y \cup x) \quad \text{for every } x \in X. \quad (5)$$

Let $\overline{O} := O \setminus (X \cup Y)$.

We construct a two-agent economy (N, ω, P) as follows. Let $N = \{1, 2\}$ and assume agent 1 reports the preference relation P_1 . Agent 2 reports a lexicographic preference $P_2 \in \mathcal{L}_2$ that orders individual objects by placing all objects in $\overline{O}$ at the very top, followed by object x^* , then all objects in $X \setminus \{x^*\}$, and finally placing all objects in Y at the very bottom. Formally:

1. for all $z \in \overline{O}$ and $o \in O \setminus \overline{O} = X \cup Y$, $z P_2 o$; and
2. for all $o \in X \cup Y$, $x^* R_2 o$; and
3. for all $y \in Y$ and all $o \in O \setminus Y$, $o P_2 y$.

Define the endowment $\omega = (\omega_1, \omega_2)$, where

$$\omega_1 = Y \cup \{x^*\} \quad \text{and} \quad \omega_2 = O \setminus \omega_1 = \overline{O} \cup (X \setminus x^*).$$

Both endowments are nonempty because $|X| \geq 3$, and they partition O . We show that no balanced allocation is simultaneously Pareto efficient and individually rational.

Step 1: The endowment ω is not Pareto efficient. Consider the allocation $\nu \in \mathcal{A}$, where

$$\nu_1 = Y \cup (X \setminus x^*) \quad \text{and} \quad \nu_2 = \overline{O} \cup \{x^*\}.$$

For agent 1, $\nu_1 P_1 \omega_1$ by (5). For agent 2, the definition of P_2 implies that $x^* = \max_{P_2}(\nu_2 \triangle \omega_2)$; since $x^* \in \nu_2 \setminus \omega_2$, it follows that $\nu_2 P_2 \omega_2$. Thus ν Pareto dominates ω . (Note that ν is not balanced because $|X| \geq 3$ and hence $|\nu_1| \geq |Y| + 2 > |\omega_1|$.)

Step 2: Every other balanced allocation fails individual rationality. Toward contradiction, let $\mu \in \mathcal{A}^{\text{BAL}} \setminus \{\omega\}$ satisfy individual rationality. Balancedness of μ means that

$$|\mu_1| = |\omega_1| = |Y| + 1 \quad \text{and} \quad |\mu_2| = |\omega_2| = |\overline{O}| + |X| - 1.$$

Since $\omega_2 \supseteq \overline{O}$, and since every individual object in $\overline{O}$ is preferred to every object outside $\overline{O}$ under P_2 , individual rationality implies $\overline{O} \subseteq \mu_2$. Therefore, $\mu_1 \subseteq X \cup Y$.

If $x^* \notin \mu_2$, then individual rationality and the definition of P_2 imply that $\mu_2 = \omega_2$ and hence $\mu = \omega$, a contradiction. Consequently, $x^* \in \mu_2$ and hence $x^* \notin \mu_1$. We need the following.

Claim 6. For all $S \subseteq X \cup Y$ with $|S| = |Y| + 1$ and $Y \not\subseteq S$, $(Y \cup x^*) P_1 S$.

Proof. Suppose otherwise. Then there is some $S \subseteq X \cup Y$ such that

$$|S| = |Y| + 1, \quad Y \not\subseteq S, \quad \text{and} \quad S P_1 (Y \cup x^*),$$

where the strict preference follows because $S \neq Y \cup \{x^*\}$. Since $Y \not\subseteq S$, we have $Y \setminus S \neq \emptyset$. Also $S \setminus Y \subseteq X$ because $S \subseteq X \cup Y$.

Define

$$Y' := Y \cap S \quad \text{and} \quad X' := (Y \setminus S) \cup X.$$

Then X' is nonempty and disjoint from Y' , and $|Y'| < |Y|$. We show that (X', Y') also satisfies (4), thereby contradicting the minimality of $|Y|$.

Let $x \in X'$. Consider three cases.

Case 1. Suppose $x \in Y \setminus S$. Let $Z := S \setminus Y$. Then $Z \subseteq X' \setminus x$ and

$$Y' \cup Z = S \ P_1 (Y \cup x^*) \ P_1 (Y' \cup x),$$

where the last strict preference follows from monotonicity of P_1 and $Y \cup \{x^*\} \supsetneq Y' \cup \{x\}$.

Case 2. Suppose $x = x^*$. Let $Z := (Y \setminus S) \cup (X \setminus x^*)$. Then $Z \subseteq X' \setminus x$ and

$$Y' \cup Z = [Y \cup (X \setminus x^*)] \ P_1 (Y \cup x^*) \ P_1 (Y' \cup x),$$

where the first strict preference follows from (5) and the second from monotonicity of P_1 and $Y \cup \{x^*\} \supsetneq Y' \cup \{x\}$.

Case 3. Suppose $x \in X \setminus \{x^*\}$. Let $Z := (Y \setminus S) \cup \{x^*\}$. Then $Z \subseteq X' \setminus x$ and

$$Y' \cup Z = (Y \cup x^*) \ P_1 (Y \cup x) \ P_1 (Y' \cup x),$$

where the first strict preference follows from (5) and the second from monotonicity of P_1 and $Y \cup \{x\} \supsetneq Y' \cup \{x\}$.

Thus, for every $x \in X'$, there exists a subset $Z_x \subseteq X' \setminus x$ with

$$[Y' \cup (X' \setminus x)] \ R_1 (Y' \cup Z_x) \ P_1 (Y' \cup x).$$

Consequently, (X', Y') is a pair of bundles satisfying (4) with $|Y'| < |Y|$. This contradicts the choice of (X, Y) . $\square$

Recall that $x^* \notin \mu_1$, $\mu_1 \subseteq X \cup Y$, and $|\mu_1| = |Y| + 1$. We consider two cases.

Case 1. Suppose $Y \subseteq \mu_1$. Then $\mu_1 = Y \cup \{x\}$ for some $x \in X \setminus \{x^*\}$. By the choice of x^* and the fact that $x^* \notin \mu_1$,

$$\omega_1 = (Y \cup x^*) \ P_1 (Y \cup x) = \mu_1.$$

Consequently, μ is not individually rational, a contradiction.

Case 2. Suppose $Y \not\subseteq \mu_1$. Then $\mu_1 \subseteq X \cup Y$ and $|\mu_1| = |Y| + 1$, so Claim 6 implies that

$$\omega_1 = (Y \cup x^*) \ P_1 \ \mu_1,$$

contradicting individual rationality of μ .

We have shown that no balanced allocation $\mu \neq \omega$ is individually rational. Hence the only balanced and individually rational allocation is ω itself. But ω is not Pareto efficient by Step 1. We conclude that no balanced allocation is both Pareto efficient and individually rational. ■

C Examples

The examples below illustrate the logical independence of the properties used in Theorems 1–4 (see Tables 1 and 2 for a summary of which properties each example satisfies on $\mathcal{L}$ and $\mathcal{R}$). Since $\mathcal{L} \subseteq \mathcal{R}$, any rule defined on $\mathcal{R}$ is also viewed as a rule on $\mathcal{L}$ by restriction.

Concretely, for each property appearing in one of these theorems, we exhibit a rule (defined on the relevant domain) that violates this property while satisfying all other properties in the theorem. To keep notation light, some examples fix convenient values for the set N of agents and the endowment ω . Analogous constructions can be given on $\mathcal{CL}$; we omit them for brevity.

Example 5. *No-trade rule* (fails IGE; satisfies MAR, BAL, WELB, IR, and SP)

Let φ^{NT} be the *no-trade rule* on $\mathcal{R}$, which selects ω for each problem. Then φ^{NT} satisfies MAR, BAL, WELB, IR, and SP on $\mathcal{R}$, but it fails IGE on $\mathcal{L}$ (and thus on $\mathcal{R}$). ◇

Example 6. *Balanced Serial Dictatorship* (fails WELB and IR; satisfies MAR, BAL, IGE, and SP)

Let φ^{BSD} be the *Balanced Serial Dictatorship* on $\mathcal{R}$, defined recursively as follows: for each $P \in \mathcal{R}$, $\varphi_1^{\text{BSD}}(P) = \max_{P_1}(O, |\omega_1|)$ and, for each $i = 2, \dots, n$,

$$\varphi_i^{\text{BSD}}(P) = \max_{P_i} \left(O \setminus \bigcup_{j=1}^{i-1} \varphi_j^{\text{BSD}}(P), |\omega_i| \right),$$

where $\max_{P_i}(X, k)$ denotes the most-preferred k -subset of X at P_i . Then φ^{BSD} satisfies MAR, BAL, IGE, and SP on $\mathcal{R}$, but it fails WELB and IR on $\mathcal{L}$ (and thus on $\mathcal{R}$). ◇

Example 7. *A rule $\varphi^{-\text{TP}}$* (fails TP and DSP; satisfies MAR, BAL, IGE, WELB, and IR)

Let $N = \{1, 2\}$, $O = \{a, b, c\}$, and $\omega = (\{a, b\}, \{c\})$. Fix $P^* \in \mathcal{L}$ with $P_1^*|_O : c, a, b$ and $P_2^*|_O : a, b, c$, and let $\mu^* = (\{a, c\}, \{b\})$. Define the rule $\varphi^{-\text{TP}}$ on $\mathcal{R}$ by

$$\varphi^{-\text{TP}}(P) = \begin{cases} \mu^*, & \text{if } P|_O = P^*|_O \\ \varphi^{\text{TPC}}(P), & \text{otherwise.} \end{cases}$$

It is easy to see that $\varphi^{-\text{TP}}$ satisfies MAR, BAL, IGE, WELB, and IR on $\mathcal{R}$.

We show that $\varphi^{-\text{TP}}$ fails TP and DSP on $\mathcal{L}$. Consider $P'_2 \in \mathcal{L}_2$ with $P'_2|_O : a, c, b$. Then $\varphi_2^{-\text{TP}}(P_1^*, P'_2) = \{a\} \setminus P_2^* \setminus \{b\} = \varphi_2^{-\text{TP}}(P^*)$, so P'_2 is a profitable manipulation for agent 2 at P^* .

Since P'_2 is both a truncation strategy and a drop strategy for P_2^* (it is obtained by dropping the singleton tail subset $\{o \in O \setminus \omega_2 \mid b R_2^* o\} = \{b\}$), $\varphi^{-\text{TP}}$ fails TP and DSP on $\mathcal{L}$ (and thus on $\mathcal{R}$). $\diamond$

Example 8. A rule $\varphi^{-\text{BAL}}$ on $\mathcal{L}$ (fails BAL; satisfies PE, WELB, IR, and SDSP)

Let $N = \{1, 2\}$, $O = \{a, b, c\}$, and $\omega = (\{a, b\}, \{c\})$. Consider the lexicographic profile P^* with $P_1^* : c, a, b$ and $P_2^* : a, b, c$, and let $\mu^* := (\{c\}, \{a, b\})$. Define the rule $\varphi^{-\text{BAL}}$ on $\mathcal{L}$ by

$$\varphi^{-\text{BAL}}(P) = \begin{cases} \mu^*, & \text{if } P = P^* \\ \varphi^{\text{TTC}}(P), & \text{otherwise.} \end{cases}$$

It is easy to see that $\varphi^{-\text{BAL}}$ satisfies PE, WELB, and IR.

We show that $\varphi^{-\text{BAL}}$ satisfies DSP (and thus SDSP by Lemma 2). Fix $P \in \mathcal{L}$, agent $i \in N$, and a drop strategy $P'_i \in \mathcal{D}_i(P_i) \setminus \{P_i\}$. Denote $P' := (P'_i, P_{-i})$ and consider three cases.

- Case 1. Suppose $P \neq P^*$ and $P' \neq P^*$. Then $\varphi^{-\text{BAL}}(P) = \varphi^{\text{TTC}}(P)$ and $\varphi^{-\text{BAL}}(P') = \varphi^{\text{TTC}}(P')$, so the deviation is not profitable because φ^{TTC} satisfies SDSP on $\mathcal{L}$.
- Case 2. Suppose $P = P^*$. If $i = 2$, then at P_2 the assignment $\varphi^{-\text{BAL}}(P) = \mu_2^*$ is agent 2's most-preferred feasible bundle (i.e., among all nonempty, proper subsets of O), so no deviation can be profitable. If $i = 1$, then any proper drop strategy for P_1 must drop object $c \in O \setminus \omega_1$, giving $P'_1 : a, b, c$. Thus $\varphi_1^{-\text{BAL}}(P) = \{c\} P_1 \{a, b\} = \varphi_1^{-\text{BAL}}(P')$.
- Case 3. The case $P' = P^*$ is impossible. Indeed, under P_1^* the bottom object is $b \in \omega_1$, so P_1^* is not a proper drop strategy for any $P_1 \in \mathcal{L}$. Similarly, under P_2^* the bottom object is $c \in \omega_2$, so P_2^* is not a proper drop strategy for any $P_2 \in \mathcal{L}$. $\diamond$

Example 9. A rule $\varphi^{-\text{WELB}}$ on $\mathcal{L}$ (fails WELB; satisfies BAL, IGE, IR, and SDSP)

Let $N = \{1, 2\}$, $O = \{a, b, c, d\}$, and $\omega = (\{a, b\}, \{c, d\})$. Consider the lexicographic profile P^* with $P_1^* : c, a, b, d$ and $P_2^* : a, b, c, d$, and let $\mu^* = (\{c, d\}, \{a, b\})$. Define $\varphi^{-\text{WELB}}$ on $\mathcal{L}$ by

$$\varphi^{-\text{WELB}}(P) = \begin{cases} \mu^*, & \text{if } P = P^* \\ \varphi^{\text{TTC}}(P), & \text{otherwise.} \end{cases}$$

It is easy to see that $\varphi^{-\text{WELB}}$ satisfies BAL, IGE, and IR. Furthermore, $\varphi^{-\text{WELB}}$ fails WELB at P^* because $d \in \varphi_1^{-\text{WELB}}(P^*)$ although $\min_{P_1^*}(\omega_1) = b P_1^* d$.

We now show that $\varphi^{-\text{WELB}}$ satisfies SDSP (hence TP and DSP). Fix $P \in \mathcal{L}$, agent $i \in N$, and a subset-drop strategy $P'_i \in \mathcal{S}_i(P_i) \setminus \{P_i\}$. Denote $P' := (P'_i, P_{-i})$ and consider three cases.

- Case 1. Suppose $P = P^*$. If $i = 2$, then $\varphi_2^{-\text{WELB}}(P) = \mu_2^*$ is agent 2's most-preferred two-object bundle at P_2^* , so no deviation can be profitable. If $i = 1$, then any proper subset-drop strategy for P_1 must drop object $c \in O \setminus \omega_1$ (possibly together with d),

which means that P'_1 is either $P_1^1 : a, b, d, c$ or $P_1^2 : a, b, c, d$. In each case, φ^{TTC} assigns $\varphi^{\text{TTC}}(P') = \omega$. Thus $\varphi_1^{\neg\text{WELB}}(P) = \{c, d\} P_1 \{a, b\} = \varphi_1^{\neg\text{WELB}}(P')$.

Case 2. Suppose $P' = P^*$. Then necessarily $i = 1$ and $P_2 = P_2^*$ (since P_2^* ranks an object $d \in \omega_2$ at the very bottom, it is not a proper subset-drop strategy for any $P_2 \in \mathcal{L}_2$). Moreover, $P'_1 = P_1^*$ can be obtained from P_1 only by dropping object d , which means that P_1 is either $P_1^1 : d, c, a, b$, $P_1^2 : c, d, a, b$, or $P_1^3 : c, a, d, b$. In each case, φ^{TTC} assigns $\varphi^{\text{TTC}}(P) = \mu^* = \varphi^{\neg\text{WELB}}(P')$, so the deviation is not profitable.

Case 3. Suppose $P \neq P^*$ and $P' \neq P^*$. Then $\varphi^{\neg\text{WELB}}(P) = \varphi^{\text{TTC}}(P)$ and $\varphi^{\neg\text{WELB}}(P') = \varphi^{\text{TTC}}(P')$, so the deviation is not profitable because φ^{TTC} satisfies SDSP on $\mathcal{L}$. $\diamond$

Example 10. A rule $\varphi^{\neg\text{MAR}}$ on $\mathcal{R}$ (fails MAR; satisfies IGE, WELB, IR, and TP)

Let $N = \{1, 2\}$ and $\omega = (\{a, b, d\}, \{c, e\})$. Fix an allocation $\mu^* := (\{b, c, e\}, \{a, d\})$. Let

$$\mathcal{R}^* = \{P^* \in \mathcal{R} \mid P_1^*|_O : a, e, b, c, d \text{ and } P_2^*|_O : a, c, b, d, e\}.$$

Since φ^{TTC} is marginal, for each $P \in \mathcal{R}^*$ we have $\varphi^{\text{TTC}}(P) = (\{a, d, e\}, \{b, c\})$.

Define the rule $\varphi^{\neg\text{MAR}}$ on $\mathcal{R}$ by

$$\varphi^{\neg\text{MAR}}(P) = \begin{cases} \mu^*, & \text{if } P \in \mathcal{R}^* \text{ and } \mu^* \text{ Pareto dominates } \varphi^{\text{TTC}}(P) = (\{a, d, e\}, \{b, c\}), \\ \varphi^{\text{TTC}}(P), & \text{otherwise.} \end{cases}$$

One easily checks that $\varphi^{\neg\text{MAR}}$ satisfies IGE, WELB, and IR (and BAL).

We show that $\varphi^{\neg\text{MAR}}$ satisfies TP. Fix $P \in \mathcal{R}$ and $i \in N$. Let $P'_i \in \mathcal{T}_i(P_i)$ and denote $P' := (P'_i, P_{-i})$. Denote the unique agent in $N \setminus \{i\}$ by j . We consider two cases.

Case 1. Suppose $P \in \mathcal{R}^*$. Consider two further subcases.

Subcase i. Suppose $\varphi_j^{\text{TTC}}(P) P_j \mu_j^*$. Then μ^* does not Pareto dominate $\varphi^{\text{TTC}}(P)$ at P , so the definition of $\varphi^{\neg\text{MAR}}$ implies that $\varphi_i^{\neg\text{MAR}}(P) = \varphi^{\text{TTC}}(P)$ and $\varphi_i^{\neg\text{MAR}}(P') = \varphi^{\text{TTC}}(P')$. Thus TP for φ^{TTC} implies that P'_i is not a profitable manipulation.

Subcase ii. Suppose $\mu_j^* R_j \varphi_j^{\text{TTC}}(P)$. Then the definition of $\varphi^{\neg\text{MAR}}$ implies that $\varphi_i^{\neg\text{MAR}}(P)$ is the most-preferred bundle in $\{\mu_i^*, \varphi_i^{\text{TTC}}(P)\}$ according to P_i . If $P' \in \mathcal{R}^*$, then $\varphi_i(P') \in \{\mu_i^*, \varphi_i^{\text{TTC}}(P)\}$, which implies $\varphi_i^{\neg\text{MAR}}(P) R_i \varphi_i^{\neg\text{MAR}}(P')$. If $P' \notin \mathcal{R}^*$, then TP for φ^{TTC} implies that

$$\varphi_i^{\neg\text{MAR}}(P) R_i \varphi_i^{\text{TTC}}(P) R_i \varphi_i^{\text{TTC}}(P') = \varphi_i^{\neg\text{MAR}}(P').$$

Case 2. Suppose $P \in \mathcal{R} \setminus \mathcal{R}^*$. If $P' \in \mathcal{R} \setminus \mathcal{R}^*$, then TP for φ^{TTC} implies that

$$\varphi_i^{-\text{MAR}}(P) = \varphi_i^{\text{TTC}}(P) \text{ } R_i \text{ } \varphi_i^{\text{TTC}}(P') = \varphi_i^{-\text{MAR}}(P').$$

Furthermore, it is not possible that $P' \in \mathcal{R}^*$. Indeed, $P' \in \mathcal{R}^*$ would imply $P'_i|_O \neq P_i|_O$, in which case $P'_i|_O$ is obtained from $P_i|_O$ by dropping a *nonempty* tail subset $X \subseteq O \setminus \omega_i$. In particular, this means $\min_{P'_i}(O) \notin \omega_i$, a contradiction. $\diamond$

D Declaration

During the preparation of this work the authors used Refine.ink in order to check for clarity, notation, and logical consistency. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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